Highlights

~~zgReal-Time Redesign of Public Transport Lines under Fluctuating Traffic Conditions~~A Structurally Constrained Predict-then-Optimize Framework for Public Transport Line Redesign under Fluctuating Traffic Conditions

- Existing bus lines are redesigned an hour ahead, only upon large forecast deviations.
- An explicit continuity objective limits disruption for passengers and operators.
- A predict-then-optimize loop: optimize on noisy forecasts, score on realized times.

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ABSTRACT

Public Transport (PT) lines are traditionally designed to optimize performance under nominal traffic conditions. In practice, however, operating conditions frequently deviate from the nominal ones, leading to substantial performance deterioration. Existing adaptation mechanisms typically rely on corrective and reactive interventions, such as stop-skipping, which prove insufficient under large or recurrent traffic fluctuations. In such contexts, incremental adjustments are inadequate, and a structural redesign of the network becomes necessary. This paper introduces a zgreal-time optimization methoddecision-support method, driven by short-term traffic forecasts and triggered only when the forecast-deviation measure exceeds a preset tolerance (Proposition 1), for proactively redesigning PT networks to zgmaintain high levels ofsustain user- and operator-oriented performance under pronounced traffic fluctuations. We adopt a predict-then-optimize paradigm in which PT lines are reconfigured based on predicted traffic conditions via an algorithm based on the Non-dominated Sorting Genetic Algorithm III (NSGA-III). zgTo ensure operational implementability, we explicitly constrain structural deviations by enforcing high Jaccard similarity between the baseline and redesigned networks, thereby preserving network continuity and limiting operational disruption.To promote structural continuity, we zgconstrain structural deviations by enforcing high Jaccard similarity between the baseline and redesigned networkstreat structural change as an explicit objective (Z_3), maximizing Jaccard edge overlap between the baseline and redesigned networks, and screen the initial designs by a minimum-overlap threshold, using edge overlap as a tractable proxy for limiting topology changes. To account for the effect of prediction inaccuracies on redesign quality, we construct a statistical model of the errors made by a well-established deep learning-based prediction model (Diffusion Convolutional Recurrent Neural Network) trained on real-world data. Computational results on Mandl's benchmark network, under high traffic fluctuations, show that at least 77.9% of OD pairs experience travel time reductions of at least 20%, alongside operational cost savings of up to 37%, while retaining 79% of the original network structure. Experiments on the large-scale Beijing network zgfurther demonstrate average travel-time improvements of up to 20.5%further show average travel-time improvements of up to 20.5% and operational cost reductions of 15.6%, while maintaining more than 60% route overlap with the incumbent configuration. These findings demonstrate that proactive, prediction-driven redesign can substantially improve performance, compared to current inflexible PT designs, thereby motivating a transition from current static planning paradigms toward continuous and adaptive PT network design.

1. Introduction

Public Transport (PT) networks are typically planned for nominal, scheduled conditions (Ibarra-Rojas, Delgado, Giesen and Muñoz, 2015), while real-world operations are continuously subject to variability caused by congestion and other factors (Büchel and Corman, 2020a; Danaher, Wensley, Dunham, Orosz, Avery, Cobb, Watkins, Queen, Berrebi, Connor et al., 2020), especially in mixed traffic environments. Such routine perturbations occur daily and can propagate across the network, progressively degrading service reliability (Marra and Corman, 2020). For example, in Xi'an, 74.34% of bus runs were delayed (Zhang, Xu, Lu and Fan, 2022). Beyond operational impacts, insufficiently reliable public transport can also induce behavioral responses, including increased reliance on private vehicles (Rodier, Zhang, Harold and Drake, 2025).

Existing approaches to cope with recurrent perturbations can be broadly classified as preventive, reactive, or proactive, depending on the decision horizon and intervention level. Preventive strategies enhance robustness ex ante

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through structural measures, such as dedicated bus lanes that isolate buses from mixed traffic (Russo et al., 2022). However, limited street space and high infrastructure costs often make such measures infeasible in dense urban areas (Li, Song, Li, Zhang and Miller, 2009), and preventive capacity tends to be inflexible and underutilized (International Transport Forum (ITF), 2024, 2025).

Reactive strategies attempt, at the operational level, to restore service after perturbations have occurred, for instance via stop-skipping, holding, or detouring (Peled, Lee, Jiang, Dauwels and Pereira, 2021; Tang, Ceder and Ge, 2018; Zheng, Sun, Kang, Dai and Wu, 2023). These actions deteriorate under high uncertainty, inconvenience passengers (Gkiotsalitis and Cats, 2021, §2), and are insufficient to manage cascading perturbations when no alternative network structure is available (Mahdavi and Bhouri, 2024). zgIt should be noted that proactive network redesign, as proposed in this paper, also introduces a form of user inconvenience for passengers accustomed to fixed route patterns; we explicitly manage this through the structural similarity constraint Z_3 (see §3.4.1).

A key limitation shared by both preventive and reactive strategies is that the PT network structure typically remains fixed, despite persistent changes in operating conditions. Network design decisions are revised infrequently, and traditional models assume deterministic or fixed travel times (Fonseca, 2020; Büchel and Corman, 2020b), limiting their ability to address recurrent traffic fluctuations. zg~~Proactive approaches aim to overcome this limitation by anticipating variability through short-term predictions and redesigning services in advance.~~Proactive approaches aim to overcome this limitation by anticipating variability through short-term predictions and revising the line design before the next operating interval (Peled, Lee, Jiang, Dauwels and Pereira, 2019).

The **main contribution** of this paper is to propose a new method for redesigning bus networks zg~~in real-time~~one hour ahead, updating the lines only when the forecast-trigger condition in Proposition 1 is met, in order to adapt to time-varying stochastic traffic congestion. The method follows a "predict-then-optimize" framework and provides good user- and network-centric performance, even when coupled with imperfect traffic predictions. zgThe method is meant to support human planners, not to replace them. Its primary use, and the one we adopt throughout this paper, is as an offline decision-support and policy-evaluation tool: whenever the forecast-trigger condition in Proposition 1 is met, it returns a set of candidate line designs for the coming hour and points to one balanced choice, leaving the final decision to the operator rather than rerouting buses on its own. The same machinery could in principle support near-real-time operation where routes can be changed cheaply, such as automated bus fleets, or be prepared and approved ahead of time for emergencies and planned special events (§2.3) (Peled et al., 2019); we treat these as potential deployment contexts, not as claims established in this paper.

zgConcretely, our contributions are:

(i) Problem formulation (§3): instead of designing a network from scratch, we adjust an existing set of bus lines for the hour ahead, change them only when the forecast departs enough from nominal conditions, and keep each redesign close to the network already in service; we further show that this closeness does not erode over the day, regardless of how many hours elapse (Prop. 2).

(ii) Solution method (§4): an NSGA-III-based multi-objective optimizer that balances passenger travel time, operating cost, and structural continuity, with encoding, crossover, mutation, and repair tailored to bus-line redesign.

(iii) Evaluation method (§5): the optimizer is fed the noisy travel-time predictions of a deep-learning forecaster but judged against the travel times that actually occur, so the reported figures reflect imperfect forecasts; the prediction-error distribution is fit from data via KDE.

The remainder of the paper is organized as follows. Section 2 reviews related work, Section 3 formulates the problem, Section 4 presents the solution approach, Section 5 describes the evaluation method, Section 6 reports the results, and Section 7 concludes.

2. Related Work

Previous work related to ours typically focuses on either *network design* at strategic and tactical levels, where PT lines are planned from scratch, or *reactive operational control*, which manages daily operations without altering the network structure. *Network redesign* occupies an intermediate position by adapting existing networks rather than starting anew. Within this domain, *offline redesign* relies on re-optimization for known future scenarios, whereas emerging *proactive redesign* approaches anticipate short-term uncertainty. However, these proactive methods often

overlook the critical trade-off between responsiveness and structural stability, a gap that this paper specifically addresses.

2.1. Public Transport Network Design under Nominal Conditions

The PT Network Design Problem (TNDP) consists of designing line routes and their frequencies, balancing user- and operator-centric objectives zg, and has been studied since the foundational formulations of Ceder and Wilson (1986) through successive reviews (Guihaire and Hao, 2008; Kepaptsoglou and Karlaftis, 2009; Durán-Micco and Vansteenwegen, 2022) and reference texts (Desaulniers and Hickman, 2007; Ceder, 2016). zg ~~These approaches formulate public transport networks from scratch, without reference to any pre-existing service configuration, and are intended for long-term strategic or tactical planning.~~ Most of these approaches formulate public transport networks from scratch, without reference to any pre-existing service configuration; although some formulations begin from a predefined set of candidate lines (Cervantes-Sanmiguel, Chavez-Hernandez and Ibarra-Rojas, 2023), the network topology is typically treated as a fresh design decision rather than an adaptation of an existing operational configuration. These methods are primarily intended for long-term strategic or tactical planning. The TNDP is NP-hard (Magnanti and Wong, 1984).

User-centric performance has been expressed in numerous ways. In addition to overall travel duration (John, Mumford and Lewis, 2014; Heyken Soares, Mumford, Amponsah and Mao, 2019), recent studies also examine network coverage (Cadarso, Codina, Escudero and Marín, 2017), route efficiency (Suman and Bolia, 2019), and demand fulfillment (Barahimi, Eydi and Aghaie, 2021). Operator-centric objectives are focused on limiting operational and capital expenditures (Ahern, Paz and Corry, 2022; Chen, Liu, Wang and Chen, 2018; Calabrò, Araldo, Oh, Seshadri, Inturri and Ben-Akiva, 2023), or proxies for them, such as the overall length of line routes. zg ~~Multi-objective optimization is employed to balance these different objectives, drawing on bilevel and trilevel network design frameworks and on a study of the travel-time versus fare trade-off.~~ Balancing these objectives draws on two distinct methodological strands. Multi-level formulations capture hierarchical decision-making among different agents: Szeto and Jiang (2014) established a bilevel framework in which an upper-level planner minimizes passenger transfers and a lower-level model handles PT assignment; Tian, Wang and Lin (2021) extended this to a trilevel structure with passenger routing strategies at an intermediate level. Multi-objective formulations instead seek Pareto-optimal trade-off solutions across conflicting criteria: Cervantes-Sanmiguel et al. (2023) analyzed the trade-off between travel-time reduction and fare reduction. Although both paradigms address PT network design, they respond to fundamentally different modeling questions and should not be conflated.

Note that all these methods design PT network based on nominal (e.g., average, peak, etc.) traffic conditions. Therefore, as noted by (Tan, Peng, Huang, Wang, Qiu and Liu, 2024), the resulting networks may be inefficient faced with real-time variations in traffic patterns, deviating from nominal conditions.

2.2. Reactive Operational Control under Disruptions and Uncertainty

In most real-world operations, the structure of the public transport network remains fixed as originally planned. When deviations from nominal conditions occur due to exogenous factors (Nissen, Becker, Dähne, Rabe, Scheffler, Solle and Ulbrich, 2020; Chen, Yu, Zhang and Guo, 2009), operators rely on short-term corrective actions to mitigate supply–demand mismatches. Typical examples include stop-skipping (Peled et al., 2021), short-turning, speed changing, holding (Tang et al., 2018; Manasra and Toledo, 2019; Gkiotsalitis, 2020; Olvera-Toscano, Ríos-Solís, Ríos-Mercado and Sánchez Nigenda, 2023), detouring of interrupted lines (Zheng et al., 2023), and replacement buses. Each control action has distinct disadvantages. Namely, bus holding increases the inconvenience of on-board passengers at the holding stops, short-turning requires passengers to disembark and wait for a subsequent bus trip, rescheduling affects the timetables and the crew/vehicle schedules; speed control cannot be implemented when the required speeds are beyond the safety limits, and stop-skipping results in refused passenger boardings (Gkiotsalitis, 2021). While some works integrate real-time data into operational decision-making, such as schedule adjustment for enhanced reliability (Cao, Ceder and Zhang, 2019), these frameworks focus on timetable or speed control within a fixed route structure and do not explicitly consider structural reconfiguration of the network topology under time-varying conditions. zg It should be noted that proactive network redesign, as proposed in this work, generates analogous and in some respects stronger inconveniences on the operational side, particularly affecting vehicle scheduling and driver rostering; this cost motivates our use of the structural stability constraint introduced in §3.4.1.

More importantly, when actual conditions deviate significantly from the nominal ones, such corrective actions may not be sufficient to restore performance (Guo, Araldo and Yacoubi, 2024), since the overall structure is maintained as

planned. Strategies such as stop-skipping and short-turning have also appeared in the context of offline or preplanned redesign, to generate alternative route configurations or long-term service plans under hypothetical disruption scenarios. They are outside the scope of this paper, which instead focuses on zg~~real-time line redesign~~redesigning lines for the hour ahead from short-term forecasts.

2.3. Offline and Preplanned Network Redesign under Known Scenarios

Various approaches can be employed in the bus route adjustment problem to generate adaptive bus routes, including rerouting, short-turning, skip-stopping, and branching (Cao and Ceder, 2019; Ma, Yang and Li, 2023; Wu, Kulcsár, Qu et al., 2021). In contrast to the network design approaches described in §2.1, these studies explicitly start from an existing public transport network and allow service or structural modifications under a limited set of anticipated scenarios. Previous research on bus route localized adjustments has often aimed at two main objectives: accommodating demand fluctuations and enhancing integration with subway networks.

Guan, Xiang, Wang, Yan and Zhao (2023) introduced a bilevel planning model to optimize customized bus routes offline based on a spatiotemporal state network. While effective for static planning, this method is impractical for real-time dispatching. Concretely, it relies on a computationally expensive nested algorithm and requires a full re-solve for any new event. Zhang, Jia, Bai, Mao, Ma and Zhang (2018b) investigated long-distance replacement bus services to improve coordination with urban rail networks. While effective under static demand assumptions, this approach is inherently offline and cannot be applied in real-time operational contexts. When passenger flow control is implemented in a subway system, Zhou, Hu, Huang and Deng (2021) optimized existing bus routes to transport passengers temporarily restricted from entry, in case of disruptions. Wang, Chen, Yu, An and Wu (2022) proposed a two-stage method to mitigate disruptions to bus routes under planned construction scenarios, relying on predefined work-zone impacts and static travel-time estimates. Wu, Zuo, Zhou, Wan, Zhao and Yang (2024) proposed a multi-objective ant colony algorithm to adjust existing bus routes and introduced a sequence-level similarity to prevent the redesigned network from being excessively different from the original. However, their approach remains static and cannot capture time-varying adaptations. Zheng, Sun, Wu and Kang (2024) proposed localized adaptive bus routes under anticipated disruptions, while Zhang, Xu and Wang (2024) addressed large-scale emergency disruptions assuming available spare capacity. These assumptions do not hold under recurrent traffic fluctuations, where no redundant lines exist and adaptation must occur within the same constrained topology.

2.4. Proactive Network Redesign under Predictive Uncertainty and Structural Consistency

Traditional network design handles uncertainty primarily by optimizing for expected scenarios (stochastic optimization) or worst-case bounds (robust optimization) (Bertsimas, Gupta and Kallus, 2018; Wang and Qi, 2020)zg; see Ge, Voß and Xie (2022) for a recent review of robustness and disturbances in public transport. zg~~However, these approaches typically determine a single network configuration prior to realization, without any adaptation to the realized road conditions (i.e., without any reconfiguration).~~However, these approaches typically determine a single network configuration prior to realization, not because they are unable to generate multiple configurations, but because operational contexts require stable, predictable network structures whose restructuring must be justified from an organizational standpoint (Wu et al., 2024). As a result, they provide no adaptation to the realized road conditions. Another line of work studies topological metrics to evaluate network resilience (Massobrio and Cats, 2024; Pu, Li and Ma, 2022), e.g., they measure vulnerability after a perturbation occurs. These approaches remain purely diagnostic and descriptive, rather than being prescriptive (as the approach we present in this paper), in that they do not provide guidance on the optimization of the network design.

2.5. Vehicle Routing Problems and Demand-Responsive Transport (DRT)

zg~~In Vehicle Routing Problems (VRPs) and DRT, vehicle routes are constructed on-demand and solution stability is often employed to keep revised plans close to a baseline, but this stability applies to routing or scheduling assignments rather than the structure of the PT line network itself. Moreover, DRT and other VRP-based systems do not exhibit a structured network of lines, and therefore fail to achieve sufficient passenger consolidation. By contrast, we consider a high-capacity mass transit system based on a structured network of fixed lines, reconfiguring existing lines rather than computing routes from scratch.~~Vehicle Routing Problems (VRPs) and Demand-Responsive Transport (DRT) construct vehicle routes on-demand and use solution stability concepts to limit plan revisions (Touzout, Ladier and Hadj-Hamou, 2021). However, pure DRT generally lacks the fixed-line structure of high-capacity mass transit and therefore typically does not reach the same passenger consolidation levels (Calabrò, Le Pira, Giuffrida, Inturri, Ignaccolo and Correia,

Study	Solution Method	Online/Proactive Redesign	Similarity Control	Decision Focus
(Magnanti and Wong, 1984)	Exact methods	No	No	ND
(John et al., 2014)	Multi-objective GA	No	No	ND
(Heyken Soares et al., 2019)	Scaled network GA	No	No	ND
(Cadarso et al., 2017)	Branch-and-cut	No	No	ND
(Ahern et al., 2022)	Gradient-based	No	No	ND
(Tan et al., 2024)	Congestion analysis	No	No	FS
(Guan et al., 2023)	Custom bilevel	No	No	OR
(Zhang et al., 2018b)	Tailored GA	No	No	OR
(Wang et al., 2022)	Two-stage heuristic	No	No	DR
(Wu et al., 2024)	Ant Colony	No	NW	OR
(Zheng et al., 2024)	Heuristic	No	No	DR
(Cao and Ceder, 2019)	Skip-stop heuristic	No	No	VC
(Ma et al., 2023)	Integer programming	No	No	OR
(Tang et al., 2018)	Optimal control	No	No	VC
(Manasra and Toledo, 2019)	Branch-and-bound	No	No	VC
(Gkiotsalitis, 2020)	Evolutionary	No	No	VC
(Olvera-Toscano et al., 2023)	Holding optimization	No	No	VC
(Zheng et al., 2023)	Coordination heuristic	No	No	VC
(Guo et al., 2025)	Exact methods	Partial	Penalty	NR
Our Approach	**NSGA-III based optimization**	**Yes**	**Jaccard**	**NR**

Notes: Online/Proactive Redesign – **No**: no online or proactive redesign; **Partial**: limited redesign capability (e.g., sampling-based or scenario-specific updates); **Yes**: proactive redesign enabled (e.g., predict-then-optimize). Similarity Control – **No**: no explicit similarity mechanism; **NW**: Needleman–Wunsch-based similarity; **Penalty**: penalty-based similarity; **Jaccard**: Jaccard-based similarity. Decision Focus – **ND**: from-scratch network design; **FS**: frequency setting; **OR**: offline or preplanned redesign of an existing network; **DR**: perturbation-oriented redesign or response; **VC**: vehicle or operational control; **NR**: network redesign under uncertain travel times.

Table 1
Comparative analysis of public transport optimization approaches across decision scope, online/proactive redesign capability, and similarity control mechanisms.

2022), although hybrid fixed/flexible and route-deviation services partly bridge this gap$_{zg}$, including adaptive route-choice models for integrated fixed and flexible transit (Leffler, Burghout, Cats and Jenelius, 2024). Our approach operates on a high-capacity structured line network, reconfiguring existing routes rather than computing itineraries from scratch; it is therefore conceptually distinct from VRP/DRT-based formulations.

2.6. Research gaps

Our literature review reveals several gaps in existing approaches to public transport network adaptation ($_{zg}$~~see Table 1 in the Appendix, which summarizes related work~~Table 1 summarizes related work). Most existing studies treat PT network design as a deterministic optimization problem, ignoring the inherent uncertainty and variability in urban traffic conditions. As shown in the table, only a few studies consider any form of uncertainty, and $_{zg}$~~none combine real-time network redesign capability with systematic travel time variability modeling~~few redesign the network in response to short-term forecasts while also modeling travel-time variability in a systematic way. $_{zg}$~~To the best of our knowledge, no studies have proposed a sample-based method for redesigning bus lines in real time to adjust to time-varying road conditions.~~To the best of our knowledge, few studies have proposed a sample-based method that redesigns bus lines for the hour ahead, updating them only when the forecast-trigger condition in Proposition 1 is met, under time-varying road conditions. An exception is Guo, Araldo and El Yacoubi (2025), where, however, (i) future travel times on road links were assumed to be perfectly known (which is unrealistic), so the impact of prediction errors was not considered, while we model the error of a well established prediction model, (ii) only a single simplified objective was adopted, while we solve a multiobjective optimization, and (iii) their approach is tested only on a simple hypothetical scenario, while we consider real-world dataset to fit the prediction error models.

$_{zg}$~~Our work introduces a novel methodology for redesigning routes in response to predicted time-varying road conditions through controlled modifications that balance passenger service quality and operational efficiency.~~Our work

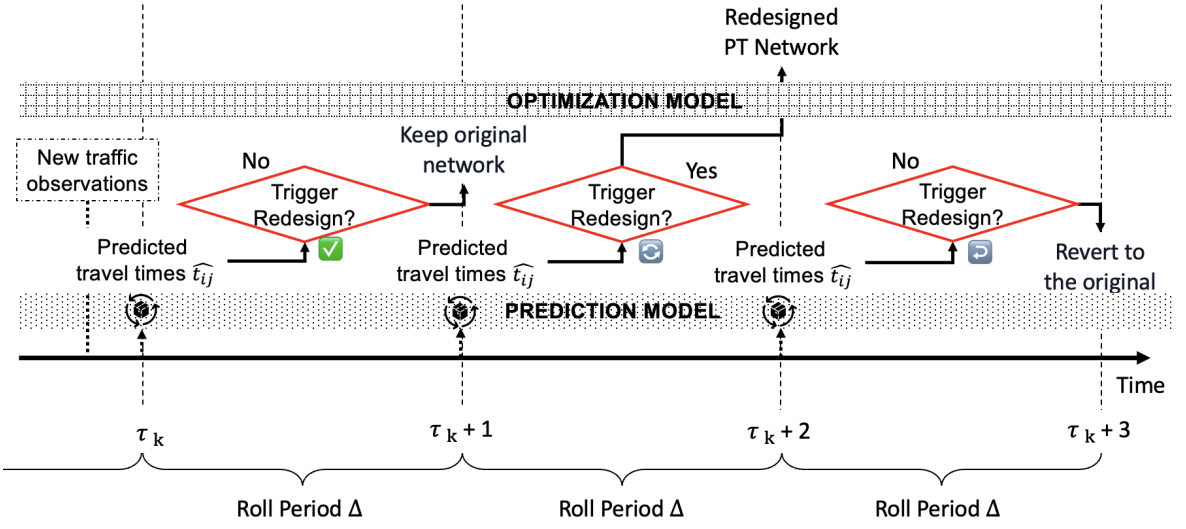


Figure 1: ~~Real-Time Network Rolling Redesign Framework.~~ Prediction-Driven Threshold-Based Line Redesign Framework.

introduces a methodology for redesigning PT lines for the hour ahead under fluctuating traffic: it uses short-term forecasts, acts only when $\bar{\Delta} > \delta_{\text{trig}}$ (Proposition 1), and balances passenger travel time against operational cost through controlled structural changes.

3. Model and Problem Formulation

As illustrated in Fig. 1, our methodology follows a predict-then-optimize framework that explicitly integrates current observations into prediction. The prediction model is first trained offline on historical data to learn travel times on road links. During operation, it receives current observations from the latest time steps and outputs a short-term prediction of link travel times $\hat{t}_{ij}$, which are then supplied to the optimization model for potential network redesign. ~~To ensure timely responsiveness, this procedure is repeated periodically, e.g., on an hourly basis, with further details provided in §3.4.1.~~ To ensure timely responsiveness, this procedure is repeated periodically, e.g., once per hour. At the start of each time slot, the new network is announced in advance, so passengers can plan a route that stays valid for their entire trip and is never changed mid-journey; we assume here that the longest trip is shorter than one time slot. This also keeps the bus routes easier to manage. Further details are given in §3.4.1. The resulting network design is then evaluated against realized travel times t_{ij} to assess the actual performance of our adaptive network under realistic conditions. The notation is summarized in Table 2.

3.1. Graph Model of PT Network

We represent the underlying road infrastructure as an undirected substrate network $\mathcal{G}^{\text{sub}} = (\mathcal{V}^{\text{sub}}, \mathcal{E}^{\text{sub}})$, where $\mathcal{V}^{\text{sub}}$ are candidate bus stops and $\mathcal{E}^{\text{sub}} = \mathcal{V}^{\text{sub}} \times \mathcal{V}^{\text{sub}}$ the potential connections among them. Observe that a single $\ell \in \mathcal{E}^{\text{sub}}$ may represent a sequence of road links from a stop to another. Let $\mathcal{L}$ denote the set of all PT lines in the network. Each PT line $\ell \in \mathcal{L}$ is defined by a set of undirected substrate edges, representing the physical stop-to-stop connections served by the line. To ensure path continuity and avoid disconnected or cyclic line configurations, the undirected edge set of each line is represented through a directed simple path graph $\mathcal{G}_\ell = (\mathcal{V}_\ell, \mathcal{E}_\ell)$ on the undirected substrate $\mathcal{G}^{\text{sub}}$, where $\mathcal{V}_\ell \subseteq \mathcal{V}^{\text{sub}}$ is the set of stops served by line ℓ, and $\mathcal{E}_\ell \subseteq \mathcal{E}^{\text{sub}}$ denotes the set of connections between consecutive stops along the path representation of line ℓ. Vehicles operating on line ℓ are assumed to traverse this path in both directions; the reverse traversal is implicitly captured through symmetric travel times (Schmid, 2014). The complete public transport network is constructed by aggregating all individual lines and is denoted by $\mathcal{G}_{\text{PT}} = (\mathcal{V}_{\text{PT}}, \mathcal{E}_{\text{PT}}, \mathcal{H})$. In this formulation, $\mathcal{V}_{\text{PT}} = \bigcup_{\ell \in \mathcal{L}} \mathcal{V}_\ell$ and $\mathcal{E}_{\text{PT}} = \bigcup_{\ell \in \mathcal{L}} \mathcal{E}_\ell$, while $\mathcal{H} = \{h_\ell \mid \ell \in \mathcal{L}\}$ represents the set of line headways, with each h_ℓ defined as the time interval between two consecutive vehicles (Vuchic, Clarke and Molinero, 1981). Accordingly, $\mathcal{G}_{\text{PT}}$ can be viewed as a line-labeled directed multigraph for modeling purposes, while operationally each line provides bidirectional service along its physical path.

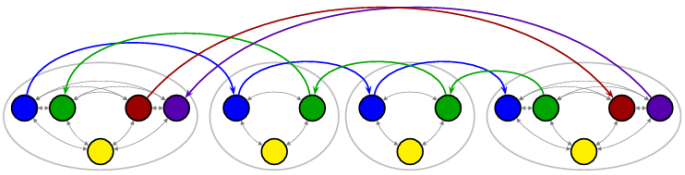


Figure 2: Example of a four-station PT network with asymmetric routes.

For each PT line $\ell \in \mathcal{L}$, headway h^ℓ can be calculated by $h^\ell = \frac{T^\ell}{n^\ell}$, where T^ℓ represents the round trip time along the entire line, and n^ℓ denotes the number of vehicles operating on that line (Eq. 4.4 of (Vuchic et al., 1981)). Let t^ℓ denote the one-way travel time from the origin terminal to the destination terminal of line ℓ, the round-trip cycle time is $T^\ell = 2t^\ell$. Accordingly, the headway of line ℓ is given by $h^\ell = \frac{2t^\ell}{n^\ell}$. We use a fixed dwell time d, at each stop to account for the time a bus spends at a stop, including the time needed for passengers to board and alight, and the time lost for accelerating and decelerating. zg~~While in practice dwell times may depend on passenger volumes, for simplicity and in line with PT network design models, we assume a constant dwell time.~~While in practice dwell times vary with boarding and alighting counts, constant dwell time is standard at the network design level (Huynh and Barthelemy, 2022; Chen, Li and Qu, 2022; Schöbel and Urban, 2022), where demand is represented as an OD matrix rather than individual trip records and the optimization horizon (one hour) far exceeds individual dwell events. Variable dwell time is primarily relevant in operational-control models (bus bunching, holding) that operate at a finer temporal resolution. We adopt this standard simplification and acknowledge it as a modeling boundary. Assuming random passenger arrivals, the average waiting time for line ℓ is given by

$$w^\ell = \frac{h^\ell}{2}, \text{ where } h^\ell \text{ denotes the service headway of line } \ell. \tag{1}$$

This assumption is standard in public transport modeling (Parbo, Nielsen and Prato, 2018; Zhang, Wang and Meng, 2018a). Liu and Miller (2020) confirm that such an assumption is appropriate when the headway is relatively small (see their § 3.3.1), and passengers simply go at the stop as soon as possible, without explicitly synchronizing at specific PT departure times; they also show that such a simple behavior is most time the most convenient for users, even better than a more complicated possible user strategies. For a transfer at stop i from line ℓ to line ℓ', the expected waiting time is therefore $w^{\ell'} = \frac{h^{\ell'}}{2}$, consistent with the waiting-time formulation in (1).

3.1.1. Virtual Node-Based Transfer Model

To model passenger route choice, we construct an auxiliary directed graph $\mathcal{G}^{\text{PAX}} = (\mathcal{V}^{\text{PAX}}, \mathcal{E}^{\text{PAX}})$ zg, referred to as the virtual-node graph; the same object is denoted $\mathcal{G}'$ in the solution algorithm of §4. The node set $\mathcal{V}^{\text{PAX}}$ comprises physical stop nodes $i \in \mathcal{V}^{\text{sub}}$ and line-specific virtual nodes v_i^ℓ for each line ℓ serving stop i. As illustrated in Fig. 2, yellow nodes denote physical stops, while colored nodes represent line-specific virtual nodes; connections between colored nodes indicate potential passenger transfers. The edge set $\mathcal{E}^{\text{PAX}}$ incorporates three distinct directed link types, each associated with a specific temporal cost. **(1) In-vehicle edges:** Connect virtual nodes v_i^ℓ and v_j^ℓ along the same line ℓ. Their weights represent in-vehicle travel times. **(2) Transfer edges:** Connect virtual nodes of different lines (e.g., v_i^ℓ to $v_i^{\ell'}$) at the same physical node. **(3) Access edges:** Connect physical nodes to virtual nodes (e.g., i to v_i^ℓ). These represent initial boarding waiting times. zg**(4) Egress edges:** Connect each line-specific virtual node back to its physical stop (e.g., v_i^ℓ to i), allowing a passenger to alight from line ℓ at stop i. They carry zero cost, since alighting incurs no additional waiting, and they ensure that every OD path terminates at the physical destination node, so that shortest paths on $\mathcal{G}^{\text{PAX}}$ from o_k to d_k are well defined. The weights of (2) and (3) correspond to passenger waiting times, which are defined in (1) in § 3.1. Transfer edges are not created at origin or destination nodes.

3.2. Stochastic Modeling of Travel Times

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For the sake of realism, we model the aforementioned bus travel times t_{ij} from stop i to stop j, for each edge $(i,j) \in \mathcal{E}^{\text{sub}}$, at a given time of day, as random variables. For simplicity, we omit explicitly specifying the time of day in the notation. Let $t_{ij}^\omega, \omega \in \Omega$ denote a realization of this variable. This is the travel time we would measure if we take a snapshot of the network at a certain point in time. Observe that, if we take

another snapshot (at the same time of day, but in another day), we would observe another travel time $t_{ij}^{\omega'}$, corresponding to another realization of variable t_{ij}. We assume that a prediction model provides predictions $\hat{t}_{ij}$ for each edge. If the prediction model were "perfect", for every sample $\omega \in \Omega$, the realizations would satisfy $\hat{t}_{ij}^{\omega} = t_{ij}^{\omega}, \forall \omega \in \Omega, (i,j) \in \mathcal{E}^{\text{sub}}$. This is obviously impossible in reality, as any realistic prediction model is subject to some errors. Let the prediction error (which is another random variable) be:

$$\epsilon_{ij} \stackrel{\text{def}}{=} \hat{t}_{ij} - t_{ij} \tag{2}$$

zgNote that ϵ_{ij} may take negative values, corresponding to cases where the model overestimates speed and therefore underestimates actual travel time. The resulting $\hat{t}_{ij} = t_{ij} + \epsilon_{ij}$ remains positive in practice: the empirically fitted error distribution (see §5.2) has negligible probability mass at values that would drive $\hat{t}_{ij}$ below zero, given the scale of nominal link travel times. whereas t_{ij} and $\hat{t}_{ij}$ denote the realized and predicted travel times, respectively. Let e_{ij}^{ℓ} indicate whether edge (i,j) is used by line ℓ. The time from one terminal to the other on line ℓ is:

$$t^{\ell} = \underbrace{\sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{N}} t_{ij}\, e_{ij}^{\ell}}_{\text{in-vehicle travel time}} + \underbrace{d \cdot \left(\sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{N}} e_{ij}^{\ell} - 1 \right)}_{\text{dwell time at stops}}. \tag{3}$$

Here, d denotes a fixed dwell time per stop. Since the actual link travel times t_{ij} for edge (i,j) are inherently uncertain, the true line travel time t^{ℓ} cannot be determined *a priori*. We can only calculate a "predicted" $\hat{t}^{\ell}$, obtained by replacing t_{ij} with $\hat{t}_{ij}$ in (3).

3.3. Trip Travel Times

Let $\mathcal{G}^{\text{sub}}$ and $\mathcal{G}_{\text{PT}}$ denote the substrate and public transport (PT) graphs, respectively (§3.1). Each OD pair $k \in \mathcal{K}$ is defined by an origin stop $o_k \in \mathcal{V}^{\text{sub}}$ and a destination stop $d_k \in \mathcal{V}^{\text{sub}}$. We assume origins and destinations are both stops, since we take responsibility of user trips from the moment they enter PT to the moment they exit, as considered in recent graph-based public transport modeling studies (Zhao, Mihăiţă, Ou, Grzybowska and Li, 2024).

Assumption 1 (Deterministic Shortest-Path Passenger Routing)**.** zg~~*Given a fixed PT network design, including line structures* $(e_{ij}^{\ell}, a_i^{\ell}, b_i^{\ell})$ *and fleet allocations* n^{ℓ}*, passengers are assumed to behave rationally and select routes that minimize their individual travel time on the designed PT network. Under this behavioral assumption with additive link costs, OD paths are induced by the design and can be computed via shortest-path algorithms.*~~*Given a fixed PT network design, including line structures* $(e_{ij}^{\ell}, a_i^{\ell}, b_i^{\ell})$ *and fleet allocations* n^{ℓ}*, each passenger is assigned to the single route that minimizes their individual travel time on the designed PT network, measured as the sum of in-vehicle, dwell, initial-waiting, and transfer-waiting costs along the path (§3.1.1). This is a deterministic, all-or-nothing shortest-path assignment: the whole demand of an OD pair follows one least-cost path, computed by a shortest-path algorithm on the virtual-node graph* $\mathcal{G}^{PAX}$*. Such deterministic shortest-path (user-optimal) routing is a standard assignment rule in transit network design (Goerigk and Schmidt, 2017; Schöbel and Urban, 2022; John et al., 2014). It abstracts away frequency-based common-line (hyperpath) route choice, in-vehicle crowding and capacity, and transfer aversion; we adopt it for tractability at the network-design level, in line with the cited transit-network-design literature.*

For any network configuration $\mathcal{G}_{\text{PT}}$ and any realization $\omega \in \Omega$, each link (i,j) can be traversed in time t_{ij}^{ω}. Based on such values, we compute shortest paths for any OD pair $k \in \mathcal{K}$ and denote the trip travel time as t_k^{ω}. Note that t_k^{ω} includes (i) in-vehicle travel times, (ii) dwell times, (iii) initial waiting times, and (iv) transfer waiting times, according to §3.1.1. Let t_{ij} be the random variable with realizations $t_{ij}^{\omega}, \omega \in \Omega$. To formalize these paths, we define binary dependent variables $p_{ij}^{k\ell,\omega}$ indicating whether arc (i,j) on line ℓ belongs to the shortest path for OD pair k in realization $\omega \in \Omega$. These variables are uniquely determined by the network state and computed algorithmically (§4) rather than being direct decision variables.

Let $\mathcal{G}_{\text{PT}}^{\text{org}}$ denote the original PT network, i.e., "master" schedule of PT over a typical day of operation. Classically, this is the network that will be operated every day. We instead allow adaptation, which means that the PT line network at a certain time is $\mathcal{G}_{PT}$, which may or may not correspond to the original one (Fig. 1). In our framework, three

Symbol	Description
Sets:	
$\mathcal{N}$	Set of nodes, indexed by $i, j \in \mathcal{N}$
$\mathcal{L}$	Set of PT lines, indexed by $\ell \in \mathcal{L}$
$\mathcal{K}$	Set of origin–destination pairs, indexed by $k \in \mathcal{K}$
$\mathcal{E}$	Set of feasible edges in the network, indexed by $(i,j) \in \mathcal{E}$
Parameters:	
$q_k \in \mathbb{N},\ k \in \mathcal{K}$	Demand for OD pair k (passengers) - §3.4.2.
$t_{ij} \in \mathbb{R}_+,\ (i,j) \in \mathcal{E};\ \ t^{\omega}_{ij}, (i,j) \in \mathcal{E}, \omega \in \Omega$	t_{ij} is the true travel time on link (i,j); this is a random variables; we denote its realizations as t^{ω}_{ij} - §3.2.
$t^0_{ij} \in \mathbb{R}_+,\ (i,j) \in \mathcal{E}$	Baseline link time (the ones referred to nominal conditions) - §3.3.
$\hat{t}_{ij} \in \mathbb{R}_+,\ (i,j) \in \mathcal{E}$	Predicted travel time on link (i,j) - §3.4.2.
$t^0_k \in \mathbb{R}_+,\ k \in \mathcal{K}$	Baseline trip travel time: travel time for OD pair k in the original network design $\mathcal{G}^{\text{org}}_{\text{PT}}$ under nominal (unperturbed) conditions - §3.3.
$o_k, d_k \in \mathcal{N},\ k \in \mathcal{K}$	Origin and destination node of OD pair k - §3.3.
$e^{\ell,0}_{ij} \in \{0,1\},\ i,j \in \mathcal{N},\ \ell \in \mathcal{L}$	Original network configuration (1 if link (i,j) is used by line ℓ) - §3.4.2
$M, d \in \mathbb{R}_+$	Large constant, Fixed dwell time
Upper-level decision variables (network design):	
Binary Variables:	
$y^{\ell},\ \ell \in \mathcal{L}$	1 if line ℓ is activated (operated)
$e^{\ell}_{ij},\ i,j \in \mathcal{N},\ \ell \in \mathcal{L}$	1 if edge (i,j) is used by line ℓ in new network
$a^{\ell}_i, b^{\ell}_i,\ i \in \mathcal{N},\ \ell \in \mathcal{L}$	1 if node i is the start/end terminal of line ℓ
Discrete or Continuous Variables:	
$n^{\ell} \in \mathbb{N},\ \ell \in \mathcal{L}$	Number of vehicles on line ℓ (fleet size)
$u^{\ell}_i \in \mathbb{N},\ i \in \mathcal{N},\ \ell \in \mathcal{L}$	Auxiliary variable for subtour elimination (MTZ): it denotes the order in which stop i appears in line ℓ.
$\hat{w}^{\ell} \in \mathbb{R}_+,\ \ell \in \mathcal{L}$	Average waiting time for line ℓ
$\hat{t}^{\ell} \in \mathbb{R}_+,\ \ell \in \mathcal{L}$	Travel time from one terminal to the other on line ℓ
Induced quantities (passenger assignment outputs):	
$p^{k\ell}_{ij} \in \{0,1\}$	1 if OD pair k traverses arc (i,j) using line ℓ
$\hat{t}_k \in \mathbb{R}_+,\ k \in \mathcal{K}$	zg~~Travel time of OD pair k~~Predicted trip travel time of OD pair k in the redesigned network $\mathcal{G}^{\text{adpt}}_{\text{PT}}$ (see §3.3)
zg$\hat{t}^0_k \in \mathbb{R}_+,\ k \in \mathcal{K}$	zgPredicted trip travel time of OD pair k on the original network $\mathcal{G}^{\text{org}}_{\text{PT}}$ under predicted link times $\hat{t}_{ij}$, i.e., the predicted cost of not redesigning (see §3.3)

Table 2
Notation for PT Network Re-Design Problem

notions of OD travel time are defined: (i) the actual travel time t_k, described in the paragraph above; we consider its realizations when evaluating the performance of our method. (ii) the predicted travel time $\hat{t}_k$, computed as the sum of the predicted link times $\hat{t}_{ij}$ along the shortest paths described in the paragraph above; we use these predicted values as input parameters of optimization (5)-(14); (iii) the baseline travel time t^0_k, defined as the OD travel time on the original network design $\mathcal{G}^{\text{org}}_{\text{PT}}$ under baseline link times $t^0_{ij}, \forall (i,j \in \mathcal{E})$. zgFor completeness, we additionally denote by $\hat{t}^0_k$ the predicted trip travel time of OD pair k on the original network $\mathcal{G}^{\text{org}}_{\text{PT}}$ under the predicted link times $\hat{t}_{ij}$; this represents the predicted cost of not redesigning, and is distinguished from $\hat{t}_k$, which is the predicted trip travel time on the redesigned network $\mathcal{G}^{\text{adpt}}_{\text{PT}}$.

3.4. Problem Statement

This research tackles the optimization of public transport lines zg~~in real time~~one hour ahead, redesigning them only when the forecast-trigger condition in Proposition 1 is met to adapt to fluctuating and unknown road conditions. The proposed optimization method has three complementary objectives: (i) minimizing user cost by reducing trip travel time deterioration due to fluctuations of substrate link times, with respect to the nominal ones (resulting in changes of actual travel time t_k experienced on OD pair k, relative to the baseline travel time experienced in the original network,

which we denote as t_k^0), (ii) lowering operator costs represented by total vehicle-hours traveled by the fleet of buses, and (iii) avoiding excessive changes in the PT network compared to the original to improve operational manageability and prevent user disorientation.

Our optimization problem addresses the following three decision problems: (a) how to dynamically modify PT lines (line topology, fleet allocation, headways) to adapt to actual road conditions when they severely deviate from the expected ones? (b) which route should we make available for users to travel among OD pairs? and (c) where to locate the terminals of each line?

3.4.1. Timeframe and Stability Considerations

Reconfiguring PT lines may affect users and operators in distinct ways.

For passengers, changes in routes can reduce perceived regularity and predictability[zg], a recognized determinant of transit network design quality (van Oort and van Nes, 2009). First, we assume that smartphone navigation apps are available to provide passengers with the best itinerary on the current PT network, so that they can navigate the reconfigured network structure. This is in line with real-world observations, where navigation apps are increasingly used (Federal Highway Administration, 2016) and more and more passengers rely on them (Durand, Hamersma, van Oort and Hoogendoorn, 2024). Ghahramani and Brakewood (2016, p 147) report that 80% of New York PT passengers use those apps twice or more in one day. Li and Shalaby (2024) show that "people seek out transit information the most during times of uncertainty" to find their best itinerary. [zg]While smartphone adoption for transit guidance is high and growing (Federal Highway Administration, 2016; Durand et al., 2024), we acknowledge that not all passengers have or regularly use smartphones. Complementary information channels, such as stop-level displays and real-time driver announcements, remain important for equitable and robust service provision; smartphone navigation is therefore treated as a predominant rather than universal mechanism. In addition to the availability of smartphone apps, behavioral trends of PT users suggest that they could be willing to adapt their trips to reconfigured network structures if this would allow them to arrive at their destination more efficiently. Indeed, empirical evidence suggests that passengers operate a trade-off between efficiency and regularity (Devarasetty, Burris and Shaw, 2012). Specifically, some travelers may prefer an alternative path that ensures on-time arrival over a habitual route that, under adverse traffic conditions on a given day, results in delay. In this sense, deviations from regular routes are not necessarily detrimental if they yield tangible gains in travel time reliability.

For the operator, frequently updated itineraries may increase cognitive load on the drivers. Nonetheless, in practice, several agencies operate services with "route deviation", in which vehicles may depart from established routes within a defined area (National Rural Transit Assistance Program (National RTAP), 2024). We assume that such operating areas have been defined ex ante for liability reasons. [zg]How demanding these assumptions (an hour of notice, predefined deviation zones, and app-based guidance for passengers) are depends on the setting, and this shows where the method is realistic today and where it fits more naturally. With automated buses there is no driver to retrain and the vehicles already reroute freely (Peled et al., 2019), so redesigning the network each hour fits well. For emergencies and planned events, the redesign covers a single corridor and is prepared beforehand, so the hour of notice is planning time rather than a real-time deadline, and approved versions can be readied in advance. Where buses still have fixed drivers, the method is best used offline, to support planning and to test what-if scenarios: it advises the planner instead of changing services on its own. This is how we use it in this paper.

Although PT line reconfiguration is implementable, PT lines are more easily manageable if disruptive reconfigurations are limited. For this reason, we measure structural stability using the Jaccard index (Jaccard, 1901), defined for the edge sets of the original and redesigned networks as

$$J(\mathcal{E}_{\text{orig}}, \mathcal{E}_{\text{new}}) = \frac{|\mathcal{E}_{\text{orig}} \cap \mathcal{E}_{\text{new}}|}{|\mathcal{E}_{\text{orig}} \cup \mathcal{E}_{\text{new}}|} \in [0, 1]. \tag{4}$$

It provides a simple, system-wide measure of overlap and has been shown to be particularly effective in sparse networks (Travieso, Benatti and Costa, 2024), which reflect typical PT topologies. Unlike sequence-based metrics such as edit distance (Cheng, Yan, Wu, Huang and Zhang, 2025), which focus on local route alignment and scale poorly in multi-line systems (Jain, Roy, Meher, Chakrabarti and De, 2024), the Jaccard index offers a tractable, network-level notion of similarity that directly matches our design objectives. We assume access to a short-term traffic prediction module, [zg]~~as is often the case in urban operations~~as is increasingly available from data-driven short-term traffic-speed forecasters deployed on urban road networks (Li, Yu, Shahabi and Liu, 2017; Liao, Zhang, Wu, McIlwraith, Chen, Yang, Guo and Wu, 2018), and focus on the subsequent optimization step. [zg]~~As shown in Fig. 1, we consider an hourly~~

~~monitoring policy: if average relative deviation $(1/|\mathcal{E}^{\text{sub}}|)\sum_{(i,j)\in\mathcal{E}^{\text{sub}}}|\hat{t}_{ij}-t^0_{ij}|/t^0_{ij}$ remain within a tolerance (e.g., 0.1), the network is kept unchanged or reset to the original configuration to avoid accumulated drift; larger deviations trigger a redesign. This selective updating strategy preserves temporal consistency while facilitating response to fluctuating traffic conditions. Moreover, deciding reconfigurations (when needed) always 1 hour in advance, gives the operator and drivers the needed time to adapt. In the hourly implementation, this one-hour advance-notice window is used as an offline planning period before the next operating interval. Because structural continuity is carried by the objective Z_3, we report the full trade-off between structural overlap and passenger benefit along the Pareto front rather than committing to a single operating point; the monitoring tolerance and the redesign trigger introduced above are the only thresholds fixed a priori.~~As shown in Fig. 1, we use the following hourly trigger.

Proposition 1 (Forecast-triggered line redesign)**.** *At the start of a time slot, define* $\bar{\Delta} \overset{\text{def}}{=} \frac{1}{|\mathcal{E}^{sub}|}\sum_{(i,j)\in\mathcal{E}^{sub}}\frac{|\hat{t}_{ij}-t^0_{ij}|}{t^0_{ij}}$*. For a monitoring tolerance* $\delta_{\text{trig}} \in [0,1]$*, keep the announced network if* $\bar{\Delta} \le \delta_{\text{trig}}$*. If* $\bar{\Delta} > \delta_{\text{trig}}$*, solve* (5)–(14) *with the predicted link times* $\hat{t}_{ij}$*; the selected solution becomes* $\mathcal{G}^{adpt}_{PT}$ *for the next one-hour slot, subject to operator approval.*

The trigger decides whether a redesign is attempted; the extent of structural change is then controlled by Z_3. The Jaccard distance inherently prevents abrupt structural jumps between redesigns. Specifically, if redesigned public transport (PT) network $\mathcal{G}^{(k)}_{\text{PT}}$ at each time interval k remains sufficiently similar to the original configuration $\mathcal{G}^{(0)}_{\text{PT}}$, the pairwise similarity between zg~~successive redesigns $\mathcal{G}^{(k)}_{\text{PT}}$ and $\mathcal{G}^{(k+1)}_{\text{PT}}$ is ensured to be bounded from below.~~any two redesigns $\mathcal{G}^{(k)}_{\text{PT}}$ and $\mathcal{G}^{(k+n)}_{\text{PT}}$ is bounded from below, regardless of the gap n.

Proposition 2 (zgHorizon-independent pairwise stability)**.** *Let* $\mathcal{E}^{(k)}$ *denote the set of active edges in the redesigned network at step k,* zg~~*and analogously for the original network and for the next redesign. Then, for any consecutive redesigns k and k+1*~~*and* $\mathcal{E}^{(0)}$ *for the original network. Then, for any two redesign steps* k *and* $k+n$ $(n \ge 1)$*, it holds that*[1]

$$J\big(\mathcal{E}^{(k)},\mathcal{E}^{(k+n)}\big) \;\ge\; J\big(\mathcal{E}^{(k)},\mathcal{E}^{(0)}\big) + J\big(\mathcal{E}^{(k+n)},\mathcal{E}^{(0)}\big) - 1.$$

3.4.2. Objective Functions

We formulate our public transport network adaptation problem as a three-objective optimization, as expressed in (5). The first objective, Z_1, minimizes passenger travel time deterioration, where q_k represents the travel demand flow. The use of $\max(\hat{t}_k - t^0_k, 0)$ guarantees that only travel-time increases relative to the nominal benchmark are penalized, thereby avoiding situations where improvements for certain OD pairs compensate for deterioration in others. The second objective Z_2 minimizes operational costs through total vehicle-hours.zgHere, operational cost is operationalized as the total vehicle-hours traveled by the fleet, a common proxy at the network-design level (Arbex and da Cunha, 2015). zgConcretely, $2\hat{t}^{\ell}$ is the round-trip running time of line ℓ and n^{ℓ} the number of vehicles assigned to it, so $Z_2 = \sum_{\ell} n^{\ell}\, 2\hat{t}^{\ell}$ aggregates fleet size and route length into a single vehicle-time figure per service cycle; it is a design-level proxy rather than a literal vehicle-hour count over the one-hour window (which would equal the active fleet multiplied by the window length), and we report its effect through the relative Operational Cost Reduction (§6.1.3). It captures the dominant fleet-operation effort but excludes crew wages, deadheading, layover, and depot costs, which depend on vehicle scheduling and driver rostering and lie outside the present scope. We retain the term "operational cost" throughout, with this vehicle-hour proxy understood. The third objective Z_3 minimizes network structural changes using the Jaccard index (§ 3.4.1), calculated as the ratio between the number of shared edges (i.e., those present both in the original and in the re-designed network) and the total number of edges (i.e., the union of the edges in the two networks). zgStructural continuity is handled as an objective (Z_3), not as a hard constraint: the threshold $J_{\min}$ acts only as a screen on the initial population (§4.2.1, criterion (iii)) and is not re-imposed after crossover or mutation, so the search explores the full overlap range and the achieved overlap is read off the Pareto front through the Route Overlap metric (§6.1.3). Throughout the paper, the objective Z_3, the Route Overlap metric, and Proposition 2 are computed on the same set of line-labelled edges, i.e. triplets $(\min(i,j), \max(i,j), \ell)$, so that the same physical link served by different lines is counted as distinct (§4.3).

[1] The proof is provided in App. A.

$$\min_{\mathcal{G}_{\mathrm{PT}}^{\mathrm{adpt}}} \begin{pmatrix} Z_1 \\ Z_2 \\ Z_3 \end{pmatrix} = \begin{pmatrix} \underbrace{\sum_{k\in\mathcal{K}} q_k \cdot \max(\hat{t}_k - t_k^0, 0)}_{\text{Passenger travel time deterioration}} \\ \underbrace{\sum_{\ell\in\mathcal{L}} n^\ell \cdot 2\hat{t}^\ell}_{\text{Total vehicle-hours (operating cost)}} \\ \underbrace{1 - \frac{\sum_{\ell\in\mathcal{L}}\sum_{(i,j)\in\mathcal{E}} \min(e_{ij}^\ell, e_{ij}^{\ell,0})}{\sum_{\ell\in\mathcal{L}}\sum_{(i,j)\in\mathcal{E}} \max(e_{ij}^\ell, e_{ij}^{\ell,0})}}_{\text{Route dissimilarity}} \end{pmatrix} \tag{5}$$

In (5), $e_{ij}^{\ell,0}$ and e_{ij}^{ℓ} denote whether line ℓ uses substrate link (i,j) in the original and the redesigned network, respectively. Observe that $e_{ij}^{\ell,0}$ are parameters of our problem, whereas e_{ij}^{ℓ} are decision variables. Observe that when solving our optimization problem, the realized link travel times t_{ij} are not available, we will thus optimize the decision variables based on prediction $\hat{t}_{ij}$ and then evaluate, ex-post in the results, the objective function evaluated using the realized link travel times t_{ij}.

zg**Variable domains.** Binary: $y^\ell, e_{ij}^\ell, a_i^\ell, b_i^\ell \in \{0,1\}$. Integer: $n^\ell \in \mathbb{Z}_{\geq 1}$, $u_i^\ell \in \{1,\dots,|\mathcal{N}|\}$. Continuous: $\hat{w}^\ell, \hat{t}^\ell \in \mathbb{R}_{\geq 0}$. The induced OD travel time $\hat{t}_k$ is a function of the design variables and the predicted link times $\hat{t}_{ij}$; it is computed algorithmically (see §4.3) and is not a free variable of the program.

3.4.3. Constraints

We now report the constraints that, together with the objective (5), formalize the upper-level network design problem. Passenger route variables are treated as induced (follower) variables and are discussed separately.

Performance-Related Constraints (1) and (3).

Line Structure Constraints

$$\sum_{i\in\mathcal{N}} a_i^\ell = y^\ell, \qquad \sum_{i\in\mathcal{N}} b_i^\ell = y^\ell \qquad \forall \ell \in \mathcal{L} \tag{6}$$

$$a_i^\ell + b_i^\ell \leq 1 \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{7}$$

$$e_{ij}^\ell + e_{ji}^\ell \leq y^\ell \qquad \forall \{i,j\} \in \mathcal{E},\ \forall \ell \in \mathcal{L} \tag{8}$$

$$\sum_{j:\{i,j\}\in\mathcal{E}} e_{ji}^\ell + a_i^\ell = \sum_{j:\{i,j\}\in\mathcal{E}} e_{ij}^\ell + b_i^\ell \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{9}$$

$$\sum_{j:\{i,j\}\in\mathcal{E}} e_{ij}^\ell \leq 1, \qquad \sum_{j:\{i,j\}\in\mathcal{E}} e_{ji}^\ell \leq 1 \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{10}$$

$$u_i^\ell - u_j^\ell + 1 \leq |\mathcal{N}|\left(1 - e_{ij}^\ell\right) \qquad \forall \{i,j\} \in \mathcal{E},\ \forall \ell \in \mathcal{L} \tag{11}$$

$$1 \leq u_i^\ell \leq |\mathcal{N}| \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{12}$$

$$n^\ell \geq y^\ell, \qquad n^\ell \leq \bar{n}\, y^\ell \qquad \forall \ell \in \mathcal{L} \tag{13}$$

$$\sum_{\ell\in\mathcal{L}} n^\ell \leq V_{\max} \tag{14}$$

Constraints (6)–(14) establish the topological structure and operational limits of the designed network. Constraints (6)–(7) synchronize line activation with terminal selection, mandating that every active line ($y^\ell = 1$) connects exactly one initial terminal to one final terminal, thereby preventing zero-length loops. Constraint (8) ensures that if a line is not activated ($y^\ell = 0$), there are no links associated to it. Otherwise, it ensures that the line is constructed in one direction (links (i,j) and (j,i) cannot be both selected) and then buses go back and forth along it (as specified in page 6). Route

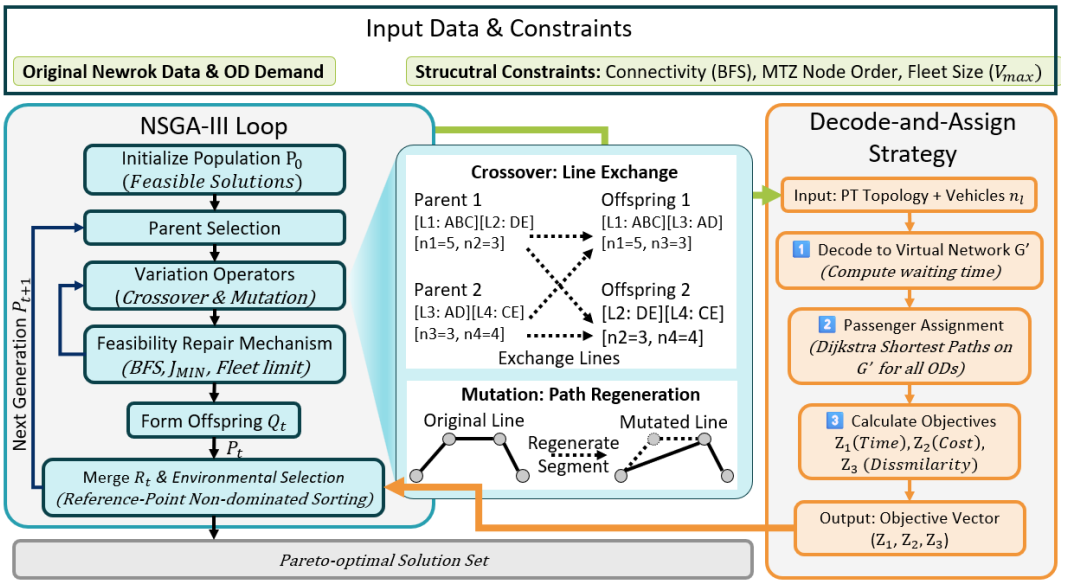


Figure 3: Flowchart of the Proposed NSGA-III-based optimization method.

continuity and linearity are enforced by the flow conservation in (9) combined with the degree Constraints in (10); the latter restrict nodes to a maximum of one predecessor and one successor, strictly imposing a non-branching simple path topology. Acyclic integrity is further guaranteed by Constraints (11)–(12), which adapt the Miller-Tucker-Zemlin (MTZ) formulation to eliminate disconnected subtours that might otherwise satisfy flow balance: if link (i, j) is used in line ℓ, then node j should appear after i along the line (its order of appearance (u_j) must be higher than i). Finally, resource allocation is governed by Constraints (13)–(14), which assign vehicles only to active lines and enforce the global fleet budget.

Induced Passenger Paths. Passenger routes are not decision variables of the network design problem. Given a fixed network design $(\mathbf{e}, \mathbf{n})$, passenger paths for each OD pair $k \in \mathcal{K}$ are deterministically induced by solving a shortest-path problem on the auxiliary passenger graph $\mathcal{G}^{\text{PAX}}$ defined in §3.1.1. Let $\hat{t}_k$ denote the resulting shortest-path travel time under predicted link costs $\hat{t}_{ij}$. For bookkeeping and interpretation, we also define $p_{ij}^{k\ell} \in \{0, 1\}$ as an induced indicator equal to one if the shortest path for OD pair k traverses arc (i, j) on line ℓ, and zero otherwise. Accordingly, $\hat{t}_k$ can be expressed as the sum of the costs of the traversed arcs (including in-vehicle, dwell, access, and transfer components) on $\mathcal{G}^{\text{PAX}}$. These induced quantities are computed algorithmically and are not part of the optimization decision set. zgThis design-evaluation decomposition is related to, but should not be conflated with, a bilevel (Stackelberg) program. Passengers do optimize their own travel time: given a fixed network, each OD demand follows a least-cost path, so a lower-level routing optimization is indeed present. What separates our setting from a Stackelberg game is that this lower-level response is non-strategic and decoupled from passenger flow. Because link travel times are exogenous to the routing decision (link costs do not depend on the passenger volumes assigned to them, i.e. there is no congestion interaction among travellers), the lower-level problem decomposes into independent shortest-path problems with a unique, deterministic solution that is computed exactly in polynomial time, rather than a flow-dependent user equilibrium requiring a fixed-point (Wardrop-type) solution. The lower-level optimum can therefore be substituted in closed form into the upper-level evaluation, collapsing the structure to a single-level combinatorial program over the design variables $(\mathbf{e}, \mathbf{n})$, whose feasibility and boundedness follow from the finite design space and the vehicle budget constraint (14). This is also what avoids the nested re-solves characteristic of strategic bilevel models.

4. Solution Algorithm

The optimization model presented in § 3 is a mixed-integer formulation featuring discrete network design choices and nonlinear objective terms. While the delay penalty in Z_1 is piecewise linear, the fractional Jaccard objective Z_3 introduces nonconvexity and global variable coupling. Since an exact linearization of this ratio would require auxiliary

constraints that severely compromise scalability, we retain the fractional structure. Given the tri-objective nature of the problem and the nonconvex trade-offs, we adopt an NSGA-III framework (Algorithm 1) to approximate the Pareto front. As illustrated in Fig. 3, we adopt a **design–evaluation decomposition** for computational tractability. The evolutionary search operates on network design variables (line topologies e_{ij}^{ℓ} and fleet allocations n^{ℓ}). For any fixed design $(\mathbf{e}, \mathbf{n})$, passenger routing is not treated as an independent optimization model; instead, it is a deterministic **decode-and-assign mapping** induced by Assumption 1. Specifically, OD travel times and the corresponding assignment outputs are computed by shortest-path calculations on the virtual graph G' (§3.1.1), which provides the objective values (Z_1, Z_2, Z_3) used by NSGA-III. Accordingly, the proposed workflow is structured into three coordinated modules: (i) **Solution Encoding and Initialization** (§ 4.1) constructs a population of structurally feasible designs; (ii) the **NSGA-III Evolutionary Loop** (§ 4.2) drives the feasible solution search, employing reference-point-based selection and variation & repair operators to evolve the topology; and (iii) the **Decode-and-Assign Strategy** (§ 4.3) performs deterministic passenger assignment to compute the objective vector (Z_1, Z_2, Z_3).

Algorithm 1 NSGA-III based Adaptive PT Network Optimization

Input: $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, $\mathcal{K}$, $\mathcal{L}$, $V_{\max}$, q_k, $\hat{t}_{ij}$, *PopSize*, *MaxIter*, N_{ref}, P_c, P_m, $J_{\min}$
Output: Approximate Pareto set $\mathcal{P}^*$

1: Initialize a population $\mathcal{P}$ of size *PopSize* via §4.1 and §4.2.1 (using parameter $J_{\min}$)
2: Evaluate (Z_1, Z_2, Z_3) for all $p \in \mathcal{P}$ via decode-and-assign (§4.3)
3: **for** $iter = 1$ **to** *MaxIter* **do**
4: Select parents from $\mathcal{P}$ via binary tournament selection
5: Generate offspring $\mathcal{Q}$ by crossover (§4.2.2) with P_c and mutation (§4.2.3) with P_m
6: Apply feasibility-oriented repair to offspring when line-level structural infeasibility (e.g., disconnection or discontinuity) is detected (Alg. 2)
7: Evaluate (Z_1, Z_2, Z_3) for all $q \in \mathcal{Q}$ via decode-and-assign (§4.3)
8: $\mathcal{R} \leftarrow \mathcal{P} \cup \mathcal{Q}$
9: Perform non-dominated sorting on $\mathcal{R}$
10: Apply NSGA-III selection with N_{ref} reference points to obtain next $\mathcal{P}$ (§4.2.4)
11: **end for**
12: **return** the non-dominated solutions in $\mathcal{P}$ as $\mathcal{P}^*$

4.1. Solution Encoding

In Alg. 1, each PT line ℓ is represented as an ordered simple path $p_\ell = (i_1, i_2, \ldots, i_{n_\ell})$ with $i_j \in \mathcal{N}$ and no repeated nodes, which enforces acyclicity by construction according to constraints (6)–(14). PT lines may either be constructed from *scratch* or obtained by *reconstructing routes from the existing network*. Starting from a current node i, the algorithm maintains a visited-node set $\mathcal{V}^{\ell}_{\text{visited}}$ and defines the set of feasible candidates as $\mathcal{N}_{\text{unvisited}}(i) = \{\, j \in \mathcal{N}(i) : j \notin \mathcal{V}^{\ell}_{\text{visited}} \,\}$, which is used whenever a new path segment needs to be generated. For lines constructed from scratch, the path is iteratively extended by selecting the next node $j \in \mathcal{N}_{\text{unvisited}}(i)$ according to the travel-time weighted probability

$$\mathbb{P}(j \mid i) = \frac{1/t_{ij}}{\sum_{k \in \mathcal{N}_{\text{unvisited}}(i)} 1/t_{ik}}, \tag{15}$$

which favors temporally efficient connections while preserving stochastic exploration. When a line is derived from the existing network, local modifications are applied by replacing a selected route segment with an alternative feasible path that preserves the original endpoints.

Simultaneously, vehicle allocation is managed by assigning integer variables n^{ℓ} to each line. During initialization (Alg. 1, line 1), each line is pre-allocated one vehicle to guarantee basic service availability. The remaining $V_{\max} - |\mathcal{L}|$ vehicles are then distributed across lines proportionally to their initial one-way travel times t^{ℓ} (computed based on nominal link travel times t_{ij}).

4.2. Evolutionary Search Procedure

Each individual encodes a complete PT network design, comprising a set of line path representations and the associated vehicle allocations. Starting from an initial population, the algorithm evolves solutions in a generational

manner. At each generation, offspring are produced through selection, variation, and evaluation, followed by the NSGA-III reference-point-based *environmental selection*, which prioritizes non-dominated solutions while maintaining diversity along the Pareto front. The overall procedure is summarized in Alg. 1.

4.2.1. Population Initialization

During population initialization (Alg. 1, line 1), each individual is generated using the line construction rules described in § 4.1. PT lines are either constructed from scratch or obtained by locally perturbing routes reconstructed from the existing network. Given a set of lines, vehicle allocations are initialized by assigning one vehicle to each active line to guarantee basic service availability. The remaining fleet is then distributed across lines according to the chromosome encoding scheme described in § 4.1, ensuring $\sum_{\ell \in \mathcal{L}} n^{\ell} = V_{\max}$.

To expedite initialization, candidate solutions are screened via a **connectivity check** on the undirected graph induced by $\mathcal{E}_{\mathrm{PT}} = \bigcup_{\ell \in \mathcal{L}} \mathcal{E}_{\ell}$, i.e., the union of all active line edges in $\mathcal{G}_{\mathrm{PT}}$. By employing a Union–Find data structure, we efficiently identify connected components, and verify that, for every OD pair $k \in \mathcal{K}$, the origin o_k and destination d_k belong to the same connected component (which implies that it is possible to go from one to the other). The procedure is repeated until *PopSize* individuals (i.e., PT network structures) are generated that satisfy: (i) structural validity (lines are simple-paths, with no branches nor cycles), (ii) connectivity-based serviceability, i.e., all OD pairs $k \in \mathcal{K}$ are connected within $\mathcal{G}_{\mathrm{PT}}$, and (iii) similarity, i.e., individual $\mathcal{G}_{\mathrm{PT}}$ is not extremely different from original PT network $\mathcal{G}_{\mathrm{PT}}^{\mathrm{org}}$, i.e., the Jaccard similarity index (4) must be above threshold $J_{\min}$. These criteria enforce a necessary (but not sufficient) condition for OD reachability; full feasibility and network performance are subsequently evaluated using the decode-and-assign objective.

4.2.2. Crossover Operation

With probability P_c, a crossover operator is applied to two parent solutions, denoted as parents A and B. Each solution consists of a fixed set of PT lines, where each line is treated as a basic inheritance unit characterized jointly by its route topology and vehicle allocation. The crossover exchanges complete lines between parents as the primary recombination mechanism; any subsequent modifications are limited to feasibility restoration.

Line inheritance is governed by a binary selection mask $\boldsymbol{\xi} = (\xi_{\ell})_{\ell \in \mathcal{L}}$, where each component is independently drawn as $\xi_{\ell} \sim \mathrm{Bernoulli}(0.5)$, so that both parents are treated symmetrically and structural diversity is promoted at the line level. Given $\boldsymbol{\xi}$, the offspring design is defined by block-wise inheritance:

$$(e_{ij}^{\ell})^{\mathrm{off}} = \xi_{\ell}(e_{ij}^{\ell})^{A} + (1-\xi_{\ell})(e_{ij}^{\ell})^{B}, \qquad \forall \ell \in \mathcal{L},\ (i,j) \in \mathcal{E}, \tag{16}$$

$$(n^{\ell})^{\mathrm{off}} = \xi_{\ell}(n^{\ell})^{A} + (1-\xi_{\ell})(n^{\ell})^{B}, \qquad \forall \ell \in \mathcal{L}. \tag{17}$$

Whenever a line topology is inherited from a parent, its vehicle allocation is inherited jointly, thereby preserving the coupling between route geometry and service frequency.

If any inherited line becomes structurally infeasible (e.g., disconnected or discontinuous), a feasibility-oriented restoration operator is applied to recover a valid simple-path representation (Alg. 2). Finally, fleet constraints ($n^{\ell} \geq 1$ for all active lines and $\sum_{\ell \in \mathcal{L}} n^{\ell} = V_{\max}$) are enforced via minimal ± 1 adjustments on a small subset of lines. This correction is a feasibility projection and is not guided by objective improvement.

4.2.3. Mutation Operation

With probability P_m, a mutation perturbs a single line ℓ via local structural modifications, including node insertion, node deletion, or segment reconnection. These operations introduce stochastic variations in line topology while preserving the overall chromosome structure. If the mutation results in a disconnected or discontinuous line, feasibility is restored by reconnecting the affected segments (Alg. 2). This repair step is feasibility-oriented and does not explicitly consider objective values. Finally, global fleet conservation ($\sum_{\ell \in \mathcal{L}} n^{\ell} = V_{\max}$) is enforced in the same manner as described in § 4.2.2.

4.2.4. Environmental Selection

After offspring generation, parent and offspring populations are merged and ranked via non-dominated sorting. Environmental selection follows the standard NSGA-III mechanism, employing a set of reference points N_{ref} to maintain population diversity (Deb and Jain, 2014).[2]

[2]The implementation details are provided in App. C.3.

4.3. Objective Evaluation via Decode-and-Assign

The evaluation phase (Alg. 1, line 7) maps each chromosome into the objective space through a systematic decode-and-assign procedure. Following Assumption 1, we treat passenger routing as a deterministic response to the network configuration, thereby decoupling routing behavior from the upper-level network design decisions. For each individual, the algorithm first constructs a comprehensive virtual-node graph $\mathcal{G}'$ that integrates in-vehicle travel times and waiting times w^{ℓ}, the latter being derived from the allocated vehicle units n^{ℓ} (§ 3.1.1). To evaluate the travel-time deterioration Z_1, the algorithm solves a shortest-path problem (Dijkstra) for every OD pair $(o_k, d_k) \in \mathcal{K}$ on $\mathcal{G}'$. The resulting shortest path yields the predicted OD travel time $\hat{t}_k$. Feasibility is enforced by construction: $\mathcal{G}'$ includes only arcs corresponding to active line segments in the decoded network design, and OD paths are initialized through access/egress connections between physical stops and their associated line-specific virtual nodes (§3.1.1). This shortest-path evaluation is performed for each candidate network, consistently reflecting the current decoded topology and headways. Simultaneously, the operational cost Z_2 and network dissimilarity Z_3 are computed. Specifically, Z_3 is quantified using a line-specific Jaccard similarity, where each network element is defined as a triplet $(\min(i,j), \max(i,j), \ell)$. This representation ensures that identical physical links served by different lines are treated as distinct entities, accurately capturing the topological shifts in the service network.

5. Evaluation Method

In order to test if our zg~~real-time~~forecast-triggered PT re-design method would perform well in a realistic situation with realistic uncertain fluctuations of link times, we need to tackle the following two challenges: (i) link travel times are inherently stochastic, and (ii) re-design decisions are based on travel time predictions, which might deviate from the true ones. To account for (i), we generate stochastic travel times for all links, with different levels of variability (§ 5.1). To account for (ii), we then generate, in § 5.2, prediction errors $\epsilon_{ij}, \forall (i,j) \in \mathcal{E}^{\text{sub}}$, statistically representative of the level of inaccuracy of well established prediction models (we consider Diffusion Convolutional Recurrent Neural Network (DCRNN) - see §5.2).

5.1. Model of Travel Time

To account for the variability in travel times, we model them as random variables, according to § 3.2. The shape we choose for this random variable is a log-normal distribution, which effectively captures the skewed nature of travel time data (Guessous, Aron, Bhouri and Cohen, 2014) and accounts for extreme delay scenarios that can occur in reality, although not often. On each link (i,j), we generate travel time realization t_{ij}^{ω} by performing Monte-Carlo sampling from random variable $t_{ij} \sim \text{log-normal}(\mu_{ij}, \sigma_{ij}^2)$ such that the median of t_{ij} corresponds to the nominal time reported in the Mandl's dataset, and the coefficient of variation (cv) matches a predetermined value: we test $cv = 0.05, 0.25, 0.5, 1.0$. The parameterization used in this sampling is stated explicitly below.

Proposition 3 (Log-normal parameterization for prescribed median and CV)**.** *Let t_{ij} be a lognormal random variable representing the travel time on link $(i,j) \in \mathcal{E}$ with median m_{ij} and coefficient of variation cv. The parameters (μ_{ij}, σ_{ij}) of the underlying normal distribution satisfy*

$$\mu_{ij} = \ln(m_{ij}), \qquad \sigma_{ij} = \sqrt{\ln(1 + cv^2)}.$$

To test the performance of our optimization when faced with different levels of hypothetical traffic variability, for each cv, we generate a set of samples Ω_{cv}, where each sample $\omega \in \Omega_{cv}$ corresponds to a collection of travel times $\{t_{ij}^{cv,\omega} \mid (i,j) \in \mathcal{E}^{\text{sub}}\}$ across the network. The proof of Proposition 3 is provided in App. B.1.

5.2. Travel Time Prediction with KDE-Based Error Modelling

zg~~Recall that our re-design method is based on predicted travel times (the input parameters of the optimization problem in § 3.4 are predicted link travel times $\hat{t}_{ij}$ rather than the true values t_{ij}, which are unknown in practice). We therefore need to evaluate whether the resulting decisions remain satisfactory despite the inevitable prediction errors $\epsilon_{ij} \overset{\text{def}}{=} \hat{t}_{ij} - t_{ij}$.~~Our optimization uses predicted link travel times $\hat{t}_{ij}$, while actual link travel times t_{ij} are unknown at decision time. We therefore evaluate whether the generated line plans remain effective under prediction errors $\epsilon_{ij} \overset{\text{def}}{=} \hat{t}_{ij} - t_{ij}$.

Recall from § 3.2 that prediction errors are modeled as random variables. We generate them through a three-stage procedure:

- In Stage 1 (see § 5.2.1), we train a prediction model to predict link travel times.
- In Stage 2 (see § 5.2.1), we use the trained model to predict new travel times and we collect the corresponding error samples.
- In Stage 3 (see § 5.2.2), we estimate the error probability distribution using the collected error samples via Kernel Density Estimation (KDE).

zg~~As for the prediction model in Stage 1, we adopted DCRNN as a well-established spatio-temporal model; the goal was to obtain realistic prediction errors rather than compare forecasting models, and the adoption of DCRNN was presented as a conservative evaluation.~~We use the Diffusion Convolutional Recurrent Neural Network (DCRNN) (Li et al., 2017) as a representative spatio-temporal predictor to generate empirical one-hour-ahead residuals. The objective is not to benchmark forecasting models, but to expose the line-plan generation method to realistic forecast errors. The dataset-specific training and residual-extraction procedure is summarized below and detailed in App. B.2.

zg

Why the residual distribution matters. For predict-then-optimize public-transport decisions, forecast-error shape matters, not only average error magnitude: bias, skewness, and heavy tails can affect downstream optimized decisions (Peled et al., 2021).

5.2.1. Stage 1 and 2: Prediction model and error generation

We train a Diffusion Convolutional Recurrent Neural Network (DCRNN). Due to the absence of historical traffic data for Mandl's Swiss network, the DCRNN is trained on the METR-LA dataset (Li et al., 2017), which contains traffic speed measurements from 207 sensors in Los Angeles County collected at 5-minute intervals over four months. zg~~Although METR-LA is a different network from Mandl's, we apply the error patterns derived via DCRNN in the METR-LA to our Mandl's network, as a practical way to reproduce the realistic errors of predictive models. This approximation does not affect the methodological validity of the presented approach, which focuses on optimization under uncertainty rather than on predicting models' accuracy.~~Because METR-LA is not Mandl's network, this transfer should be interpreted as a proxy error model for testing optimization under forecast uncertainty, not as evidence of network-specific forecasting accuracy on Mandl's benchmark. For the real-world Beijing case study, instead, we utilize the Q-Traffic dataset (Liao et al., 2018), which provides traffic speed and road network data for the 6th Ring Road in Beijing, China.

Post-training, we collect the 1-hour-ahead speed predictions in each link $(i, j) \in \mathcal{E}$ and the corresponding errors ϵ_{ij}.[3]

5.2.2. Stage 3: KDE modeling of prediction errors

zg~~Following the previous framework, we modelled travel-time prediction errors as stochastic inputs to the network optimization problem. Earlier text contrasted pre-specified parametric noise distributions with a data-driven, non-parametric KDE approach.~~Following Peled et al. (2021), we treat prediction errors as stochastic inputs to the network optimization problem. Unlike approaches that prescribe parametric noise distributions, we estimate the empirical residual distribution by KDE. zg~~In particular, instead of assuming a Gaussian or Weibull prediction error shape, we obtained prediction errors from an actual predicting model and fitted their distribution using KDE; the appendix detailed the implementation and reported that KDE performed best.~~This lets us model residuals from an actual predictor without imposing a Gaussian or Weibull shape (Chen, 2017; Akritas and Van Keilegom, 2001). Implementation details and distribution-fit comparisons are provided in App. B.3.

zgThe residuals are not recentred, preserving any mild empirical bias from the trained model. They are sampled independently from a pooled link-level distribution, so spatial heterogeneity and spatio-temporal correlation are not modelled; see App. B.2.1 for details.

5.2.3. Simulating predicted travel times

For each simulation realization $\omega \in \Omega$ and link $(i, j) \in \mathcal{E}^{\text{sub}}$, we implement a two-step process: (1) sample actual travel time t_{ij}^{ω} from the log-normal distribution of § 5.1, and (2) generate prediction error ϵ_{ij}^{ω} from the estimated error

[3] Training details and error collection methodology are provided in App. B.2.

distribution (20). The resulting predicted travel time is:

$$\hat{t}_{ij}^{\omega} = \underbrace{t_{ij}^{\omega}}_{\S\,5.1} + \underbrace{\epsilon_{ij}^{\omega}}_{\S\,5.2} \tag{18}$$

6. Numerical Results

zg~~We evaluate the proposed adaptive public transport (PT) redesign framework on both benchmark and real-world networks.~~We evaluate the proposed framework on Mandl's benchmark network (§6.2) and a real-world Beijing network (§6.3). In all experiments, the optimizer receives predicted link times $\hat{t}_{ij}$ and produces a redesigned network; performance is then scored against the realized travel times t_{ij}^{ω}, so that reported figures reflect decision quality under imperfect forecasts.

6.1. Experimental setup and metrics

Mandl's Swiss network, initially used by Mandl (1980), consisting of 15 nodes, 21 links, and a symmetric demand of 15,570 passenger trips, is employed as a canonical benchmark for comparison with existing PT network design methods (Liang, Xu and Wang, 2025; Ahern et al., 2022). To further assess the method's applicability under realistic operating conditions, we consider a real-world urban public transport network in a region in Beijing, China, recently analyzed by Wu et al. (2024), which comprises 301 bus stops, 3,758 road sections, and 29 bus lines. Its origin–destination demand matrix is constructed from smart card data collected during peak hours by aggregating trips between boarding station i and alighting station j.

Key parameters are set as follows: a fixed dwell time of $d = 1$ min per stop is assumed following Chen et al. (2022). zgThe forecast-trigger tolerance in Proposition 1 is set to $\delta_{\text{trig}} = 0.1$, i.e., redesign is triggered when the mean relative deviation of predicted link times from nominal link times exceeds 10%. The maximum fleet size ($V_{\max}$) is capped at 60 for Mandl's network, which permits relatively short headways (approximately 3–4 minutes in the resulting solutions) and enables the emergence of express-like services. For the Beijing network, $V_{\max}$ is set to 1,000 to accommodate a constant 12-minute headway across this large PT system.

6.1.1. Compared networks and baselines

We wish to show the benefits of adapting PT network design zg~~in real-time~~for the hour ahead, and only when the forecast-trigger condition in Proposition 1 is met, to varying road conditions via our method, zg~~against current network design approaches, which all keep the PT structure unchanged~~against selected static baselines, which keep the original line structure fixed.

For Mandl's Swiss network, the current network design approaches are represented by two networks:

- A network $\mathcal{G}_{\text{PT}}^{\text{static}}$, generated by jointly optimizing passenger travel time (Z_1) and operational cost (Z_2) via our solution method (§ 3.4) without adaptive mechanisms (therefore the network is static).
- A state-of-the-art network design $\mathcal{G}_{\text{PT}}^{\text{ver}}$, derived from the exact formulation of Vermeir, Engelen, Philips and Vansteenwegen (2021).

To ensure comparability and avoid excessively short routes, for all methods we consider $|\mathcal{L}| = 3$ lines in the Mandl's network, which results in headways of about 3-4 minutes in the obtained solutions.

For the real-world Beijing network, instead, the current design approaches are represented by

- The original bus line network, denoted by $\mathcal{G}_{\text{PT}}^{\text{orig}}$, serving as the current operational standard against which the adaptive redesign is assessed.

Across all experiments, the baseline travel time for an OD pair $k \in \mathcal{K}$, denoted by t_k^0 (§3.3), is computed as the shortest-path travel time between origin o_k and destination d_k in either network $\mathcal{G}_{\text{PT}}^{\text{static}}$, $\mathcal{G}_{\text{PT}}^{\text{ver}}$, or $\mathcal{G}_{\text{PT}}^{\text{orig}}$, depending on which one will be used as reference for comparison.

All experiments are conducted on a server equipped with two Intel Xeon Gold 5220R CPUs (48 cores, 96 threads, up to 4.00 GHz) with a maximum computational time of 1h per instance, zg~~which is compatible with the decision-making timings~~which we use here as an upper bound for our offline computational experiments within the hourly decision

cycle (see Fig. 1 and §3.4.1).[4] *(To respect double-blind review requirements, all problem instances and source code will be released will be released in a public repository upon acceptance.)*

6.1.2. Scenarios and evaluation protocol

We assess the effectiveness of our method under varying degrees of travel time uncertainty through four coefficient of variation levels:

- Minimal Variation ($cv = 0.05$)
- Low Variation ($cv = 0.25$)
- Moderate Variation ($cv = 0.50$)
- High Variation ($cv = 1.00$).

zg~~To ensure statistical reliability, we evaluate the method's robustness across multiple realizations of link travel times and prediction errors, using 5 random seeds (each corresponding to a single $\omega \in \Omega$ - §3.2).~~We evaluate the method across ten realizations of link travel times and prediction errors, using 10 random seeds (each corresponding to a single $\omega \in \Omega$ - §3.2), and quantify the resulting scenario-to-scenario uncertainty with standard deviations computed as described in Appendix D. Following our predict-then-optimize framework, the optimization phase utilizes predicted travel times $\hat{t}_{ij}$ to design adaptive networks, while the evaluation phase assesses actual performance using realized travel times $t^{\omega}_{ij}, \omega \in \Omega$. This separation enables us to quantify the impact of the prediction error on the quality of network adaptation.

6.1.3. Performance metrics

To evaluate the adaptive network under uncertain travel times, we report three application-specific performance indicators aligned with the objectives in (5):

- **20% Improvement Share (%) (IS_{20}):** The percentage of OD pairs that achieve a travel time reduction of at least 20% compared to the original PT network.
- **Operational Cost Reduction (OCR):** percentage reduction in total vehicle-hours operated relative to the original PT network (the operational-cost proxy defined in §3.4.2).
- **Route Overlap (RO):** Jaccard-based percentage overlap (4) between the edge sets of the original and adaptive PT networks.

All three metrics depend on realized travel times and are therefore random variables; in the numerical results, we report their empirical means (and zg~~95% confidence intervals) over five independent realizations~~standard deviations) over ten independent realizations.[5] zgWe distinguish the objective from the reporting metric: the optimization objective Z_1 is demand-weighted by q_k (§3.4.2), whereas IS_{20} counts OD pairs without demand weighting and therefore answers "for how many OD relations does service improve," not "how many passenger-trips benefit." The aggregate, demand-weighted passenger benefit is captured separately by the Avg. Travel Time Improvement reported for the Beijing network (§6.3, Table 4), so that both the optimization and the reported benefit reflect the distributional, demand-driven question of which travellers gain from the redesign (Fan and Machemehl, 2011; Pereira, Schwanen and Banister, 2017).

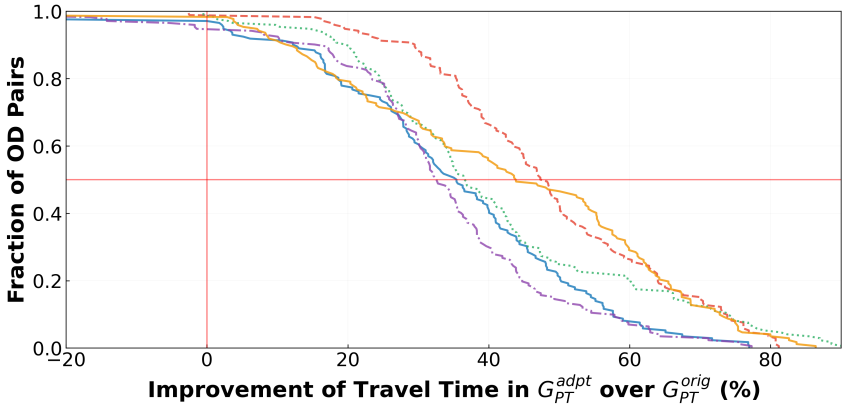
6.2. Main performance results on Mandl's benchmark

To select representative compromise solutions for the Mandl network under fluctuating travel-time, we compute the solutions on the Pareto front via the proposed Alg. 1. Then, we compute the Euclidean distance from each of those solutions to an ideal point representing the median observed values in the Pareto front, across the three objectives. The three solutions minimizing this distance are then chosen as the final candidates and are denoted as S1, S2, S3.

We use predicted travel times $\hat{t}_{ij}$ in Eq. (18) as input to the optimization problem, and we evaluate the resulting adaptive network against realized link travel times and prediction errors across multiple scenarios. Fig. 4 shows

[4] The values of the parameters of the proposed Alg. 1, as well as a thorough performance comparison against other multiobjective optimization methods, are extensively discussed in App. C.

[5] Formal definitions are provided in App. D.



Percentage of OD pairs with a trip time reduction of at least 20%

in G_{PT}^{adpt} over G_{PT}^{orig}

Realization	% of ODs
1	77.9
2	94.8
3	90.1
4	83.7
5	79.7

Figure 4: Inverse Cumulative Distribution of OD Pair Travel Time Improvement (Left) and the percentage of OD pairs with ≥ 20% reduction (Right) under High Variability (CV=1.0). The different curves correspond to 5 different random realizations $\omega \in \Omega$.

Metric	CV=0.05			CV=0.25		
	S1	S2	S3	S1	S2	S3
20% improvement share (%)	4.65 ± 1.16	5.81 ± 0.00	4.94 ± 1.35	14.34 ± 1.78	16.67 ± 1.74	13.57 ± 1.35
Op. Cost Red. (%)	10.48 ± 1.42	12.07 ± 1.59	14.37 ± 1.03	11.01 ± 0.77	10.19 ± 0.20	10.97 ± 0.36
Route Overlap (%)	82.24 ± 2.80	84.24 ± 1.47	87.04 ± 0.46	88.86 ± 0.03	85.99 ± 3.18	87.85 ± 1.78

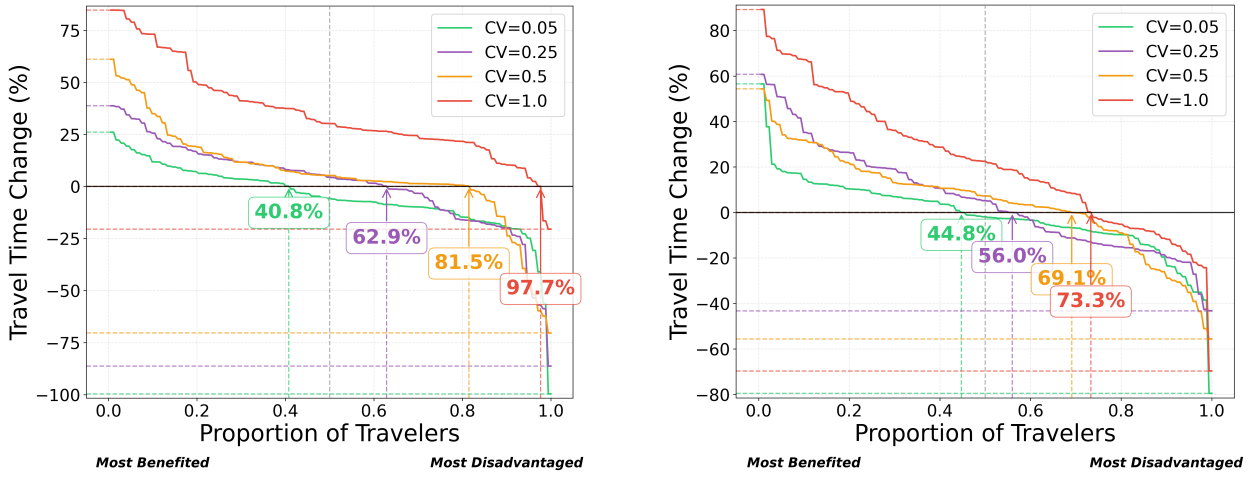
Metric	CV=0.50			CV=1.00		
	S1	S2	S3	S1	S2	S3
20% improvement share (%)	41.47 ± 1.74	42.25 ± 2.73	39.46 ± 1.49	88.18 ± 7.75	88.37 ± 8.20	88.57 ± 7.94
Op. Cost Red. (%)	16.88 ± 3.68	15.87 ± 3.95	17.18 ± 5.64	47.32 ± 2.18	44.22 ± 2.91	45.19 ± 2.64
Route Overlap (%)	91.92 ± 1.75	90.91 ± 1.38	90.99 ± 2.91	85.02 ± 0.29	84.81 ± 0.56	84.81 ± 0.56

20% improvement share. ($IS^{20\%}$): fraction of OD pairs whose travel-time improvement is at least 20%; Op. Cost Red. (OCR): Operational Cost Reduction (reduction in total vehicle-hours; see §3.4.2). Entries report mean ± standard deviation over 10 realizations.

Table 3
Performance comparison of the three solutions $S1$, $S2$, and $S3$ under different demand variability levels.

zg~~remarkable improvements of OD pair travel times~~pronounced reductions in OD-pair travel times for five of the ten realizations, shown for visual clarity, under high variability ($cv = 1.0$). Even with lower variability, improvements are significant. Table 3 summarizes average performance across zg~~five~~ten independent realizations. As expected, when variability is $cv = 0$, the central reference point (described in the previous paragraph) corresponds exactly to $\mathcal{G}_{\text{PT}}^{\text{static}}$. As travel-time variability increases, both 20% Improvement Share (IS_{20}) and Operational Cost Reduction (OCR) rise systematically, reflecting the increasing value of adaptive redesign under highly fluctuating conditions. At low variability ($cv = 0.05$), a sizable IS_{20} is still observed. This is because extreme link travel times are still possible in reality, even at low cv (the possibility of such extreme values is preserved by our choice of a Log-normal distribution §5.1. For a small-scale scenario like Mandl's network, such extremes inflate original trip times $t_k^0, k \in \mathcal{K}$. Accordingly, by exploring alternative trade-offs within the same objective space, the proposed approach can reduce passenger travel time while maintaining or improving operational costs, even in near-deterministic settings. Across all variability levels and redesign schemes, Route Overlap (RO) remains high, indicating that these improvements result from targeted adjustments rather than wholesale restructuring.

zgThese results also clarify why the compromise selection is non-trivial. The three objectives pull in different directions: reducing passenger travel-time deterioration (Z_1) can require more structural change (higher Z_3, observed as lower RO), while reducing vehicle-hours (Z_2) can shorten service and reduce passenger benefit. Thus $S1$–$S3$ should be read as balanced Pareto compromises rather than single-objective optima; a (Z_1, Z_2) projection of the static-case front is shown in Fig. 6 (Appendix C).



(a) $\mathcal{G}_{\text{PT}}^{\text{adpt}}$ vs $\mathcal{G}_{\text{PT}}^{\text{static}}$

(b) $\mathcal{G}_{\text{PT}}^{\text{adpt}}$ vs $\mathcal{G}_{\text{PT}}^{\text{ver}}$ (Vermeir et al., 2021)

Figure 5: Trip Time Distribution Comparison Under Various Variability Conditions in a randomly chosen realization.

6.2.1. Trip Travel Time Distribution Analysis

In the following sections, we will show the performance of solution $S1$. To provide a conservative performance evaluation, we considered the "worst" realization, i.e., the seed that yielded the worst results in terms of the metrics of Table 3. Fig. 5 illustrates trip time distributions across varying coefficients of variation (cv). It highlights the benefits of our adaptive public transport planning method, as travel time uncertainty increases from minimal to high levels, against current network design approaches, represented by the original static network $\mathcal{G}_{\text{PT}}^{\text{static}}$ and Vermeir's network $\mathcal{G}_{\text{PT}}^{\text{ver}}$, in accordance to §6.1.1. At low variability levels (cv=0.05), less than 50% of OD pairs improve their travel time. This occurs because these static networks are already well-optimized for stable conditions, leaving little room for enhancement. Moreover, our optimization operates on predicted travel times that inevitably include errors. This is why, as illustrated in Fig. 1, we would not trigger any reconfiguration, faced with minor fluctuations in link travel times, such as those experienced with $cv = 0.05$.

However, as variability increases, the benefits of our approach ($\mathcal{G}_{\text{PT}}^{\text{adpt}}$) become pronounced. For instance, under high uncertainty (cv=1.0), our adaptive network $\mathcal{G}_{\text{PT}}^{\text{adpt}}$ improves by at least 30% the travel time of at least 40% of zg~~travelers~~OD pairs compared to both $\mathcal{G}_{\text{PT}}^{\text{static}}$ and $\mathcal{G}_{\text{PT}}^{\text{ver}}$. zg~~This confirms the superiority of our proactive network redesign against current approaches, all keeping the network unchanged, in particular in highly fluctuating conditions.~~This suggests the potential benefit of our proactive network redesign over the selected static baselines, which keep the network unchanged, in particular in highly fluctuating conditions.

6.3. Large-scale validation on the Beijing network

We now move the analysis to the real Beijing network. To identify a representative solution for analysis, among the three solutions obtained as in Section 6.2, we only select here $S1$, i.e., the closest to the ideal point.

In addition to the previous metrics, we also compute the following

- **Avg. Travel Time Improvement (Avg.TTI):** percentage reduction in aggregate travel time across all OD pairs k relative to the original network.

$$\text{Avg.TTI} = \frac{\sum_{k\in\mathcal{K}}(t_k^0 - t_k^*)}{\sum_{k\in\mathcal{K}} t_k^0} \times 100\%, \text{ where } t_k^0, t_k^* \text{ are original and adaptive travel times.} \tag{19}$$

The results in Table 4 show that the original network exhibits some inefficiencies: indeed, the solution of our optimization yields an 11.68% Avg. travel time improvement, even at $cv = 0$. This inefficiency can be attributed to the constraints of real-world planning, where routes often evolve through incremental expansions rather than system-wide

Coeff. of variation (CV)	Avg. Route Overlap (%)	Avg. Operational Cost Reduction (%)	Avg. Travel Time Improvement (%)
0.00	90.12	6.10	11.68
0.05	87.49	6.18	12.55
0.25	81.25	8.05	14.47
0.50	73.60	11.35	18.39
1.00	65.24	15.61	20.50

Table 4
Performance comparison under varying travel-time variability based on Beijing Network.

optimization.zgAccordingly, the $cv = 0$ improvement reflects static re-optimization of the incumbent network relative to its current design, rather than prediction-driven adaptation to travel-time variability; the incremental value of adaptation is the additional gain observed as cv increases. The benefit of modifying the PT lines to adapt to travel time variability emerges as variability (cv) increases: the PT lines optimized by the algorithm become more and more different than the original static ones (lower route overlap), to achieve both travel time improvements and line shortening, which in turn yields lower operational cost.

7. Conclusion

This paper introduced a novel framework for zg~~real-time PT proactive network redesign that maintains high performance under uncertain traffic conditions~~redesigning PT lines for the hour ahead from short-term forecasts, acting only when the forecast deviation from nominal conditions exceeds a preset tolerance, and keeping each redesign close to the network already in service. We formulated a multi-objective optimization problem balancing passenger travel time, operational costs, and network stability. Our predict-then-optimize approach relies on an NSGA-III-based optimization framework that proactively redesigns bus lines based on predicted line travel times.

We evaluated our approach against the inevitable prediction errors of well-established traffic-predicting models. To this aim, we fit the prediction error distribution via kernel density estimation. Experiments on Mandl's network and a real-world Beijing network demonstrate effective adaptation across varying levels of travel-time uncertainty. Results indicate that, faced with non-negligible link travel-time variability, the proposed approach zg~~yields substantial OD-level trip travel-time reductions, while achieving operational cost savings and preserving~~yields OD-level trip travel-time reductions, while achieving operational cost savings over the selected baselines and scenarios considered, and preserving a high degree of structural similarity with the baseline network. zg~~These findings highlight the importance of moving beyond static network design toward continuous, adaptive redesign in dynamic operating environments.~~Overall, the gains grow with travel-time variability and come with high route overlap, indicating that the benefit does not rely on large structural change. A natural extension is to enrich the forecast-error model with explicit spatial and temporal correlation, allowing network-wide common shocks to be represented.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The Mandl benchmark instance is publicly available in the cited literature. The Beijing network data are derived from third-party sources cited in the paper. The problem instances and source code will be released in a public repository upon acceptance.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a generative AI tool in order to improve the language and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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A. Proof and Discussion of Proposition 2

Proof. Setting $A = \mathcal{E}^{(k)}$, $B = \mathcal{E}^{(0)}$, and zg~~$C = E(k+1)$~~$C = \mathcal{E}^{(k+n)}$. The Jaccard distance between two sets X and Y is

$$d_J(X,Y) = 1 - \frac{|X \cap Y|}{|X \cup Y|} = 1 - J(X,Y).$$

Levandowsky and Winter (Levandowsky and Winter, 1971) noted that this dissimilarity measure satisfies the triangle inequality on finite sets, so that d_J may be regarded as a distance function. Hence,

$$d_J(A,C) \le d_J(A,B) + d_J(B,C).$$

Substituting $d_J(X,Y) = 1 - J(X,Y)$ into the above inequality yields

$$1 - J(A,C) \le [\,1 - J(A,B)\,] + [\,1 - J(B,C)\,],$$

which is equivalent to

$$J(A,C) \ge J(A,B) + J(B,C) - 1.$$

zg~~, gives the desired result.~~which gives the desired result. ■

Remark. This implies that if two consecutive designs exhibit a similarity of $\ge 90\%$ relative to the original (consistent with our numerical results), the similarity between these two consecutive designs is guaranteed to be at least 80%. zgThe crucial point is that, because the structural objective Z_3 is measured against the original network $\mathcal{E}^{(0)}$ at every step rather than against the preceding design, the right-hand side does not depend on the gap n. Variability therefore does not accumulate or propagate as the day progresses: the similarity between redesigns one hour apart and the similarity between redesigns twelve hours apart obey the same lower bound. While the worst-case guarantee $2J_{\min}-1$ is loose (and vacuous at $J_{\min} = 0.5$), the empirically observed similarity to the original network is much higher than $J_{\min}$: Tables 3 and 4 report route overlaps of roughly 0.85–0.90, which by the inequality above guarantees $J(\mathcal{E}^{(k)}, \mathcal{E}^{(k+n)}) \ge 0.70$–$0.80$ between any two redesigns of the day. The hourly monitoring policy (§3.4.1), which resets the network to $\mathcal{G}_{\mathrm{PT}}^{\mathrm{org}}$ whenever deviations fall within tolerance, provides a further safeguard against long-term drift.

B. Stochastic Modelling of Travel Times and Prediction Errors

B.1. Parameter Derivation for Log-Normal Distribution

Proof of Proposition 3. For a lognormal random variable $X \sim \mathrm{LogNormal}(\mu, \sigma^2)$, the median is $\mathrm{Med}(X) = \exp(\mu)$, so specifying the median m yields

$$\mu = \ln(m).$$

The mean and variance of X are

$$\mathrm{E}[X] = \exp\!\left(\mu + \tfrac{1}{2}\sigma^2\right), \qquad \mathrm{Var}(X) = \left(\exp(\sigma^2) - 1\right)\exp\!\left(2\mu + \sigma^2\right).$$

Hence the coefficient of variation is

$$\begin{aligned} cv &= \frac{\sqrt{\mathrm{Var}(X)}}{\mathrm{E}[X]} \\ &= \frac{\sqrt{\left(\exp(\sigma^2) - 1\right)\exp(2\mu + \sigma^2)}}{\exp(\mu + \tfrac{1}{2}\sigma^2)} \\ &= \sqrt{\exp(\sigma^2) - 1}. \end{aligned}$$

Equating this expression with the prescribed value of cv gives

$$cv^2 = \exp(\sigma^2) - 1 \quad \Longrightarrow \quad \sigma^2 = \ln\!\left(1 + cv^2\right) \quad \Longrightarrow \quad \sigma = \sqrt{\ln\!\left(1 + cv^2\right)}.$$

■

B.2. Traffic Speed Prediction and Error Estimation

To generate realistic stochastic prediction errors, we employ the Diffusion Convolutional Recurrent Neural Network (DCRNN) (Li et al., 2017) to model spatiotemporal traffic dependencies. Two benchmark datasets are utilized: **METR-LA** (4 months of highway data from 207 sensors) as a proxy for Mandl's network, and **Q-Traffic** (61 days of Beijing 6th Ring Road segments) for the real-world case study. We split both datasets, chronologically, as recommended by Li et al. (2017) and Lu, Lv, Xie, Du, Xiong, Sun and Wang (2022), respectively: we train the prediction model on the first part (training set), and we measure prediction errors on the last part (test set).

The conversion from predicted speed $\hat{v}(t)$ to travel-time error $\epsilon(t)$ accounts for differences in available network attributes. For **METR-LA**, where segment lengths are not directly provided, we define directed links between upstream sensors s and downstream sensors r, determining the link length d via the network distance between them. The link speed is then approximated by averaging the endpoint measurements: $\hat{v}_{(s,r)}(t) = (\hat{v}_s(t) + \hat{v}_r(t))/2$. In contrast, **Q-Traffic** explicitly provides the length d for each road segment i, and the model output directly represents the segment speed $\hat{v}_i(t)$. In both cases, the travel-time error $\epsilon(t)$ is derived as:

$$\text{time to traverse a link: } t(t) = \frac{d}{v(t)}, \quad \text{predicted time: } \hat{t}(t) = \frac{d}{\hat{v}(t)} \implies \text{prediction error: } \epsilon(t) = \hat{t}(t) - t(t).$$

We pool all empirical prediction errors on the test sets, across links and time steps, to estimate a representative error distribution via Kernel Density Estimation (KDE) (App. B.3). Finally, independent samples ϵ_{ij}^{ω} are drawn from the estimated distribution KDE for each link (i, j) and scenario ω, and the predicted link travel times $\hat{t}_{i,j}$, used to make decisions via solving (5)-(14), are computed adding ϵ_{ij}^{ω} to the actual travel time $t_{i,j}$, according to (18). We do this in both Mandl and Beijing scenarios. Under this modeling choice, all links share the same error distribution. zg ~~Moreover, because errors are sampled independently for each link and scenario, this first-order model abstracts away the spatial and temporal correlation of real forecast errors (e.g., corridor-level or incident-level common shocks). The reported results should therefore be read as an evaluation under sampled independent residual scenarios, and correlation-aware forecast-error modeling remains an important extension.~~

B.2.1. Forecast-error modelling caveats

In the numerical experiments, synthetic predictions are constructed as $\hat{t}_{ij}^{\omega} = t_{ij}^{\omega} + \epsilon_{ij}^{\omega}$, where ϵ_{ij}^{ω} is drawn from the KDE fitted to empirical DCRNN residuals. This preserves the observed residual scale and any mild empirical bias, since the residuals are not recentred. However, it differs from a genuine ex-ante forecast, which is formed before realization and has no access to t_{ij}^{ω}. Moreover, pooling residuals and sampling them independently abstracts away link-specific error distributions and spatio-temporal error correlation. The experiments therefore test the optimizer under controlled noisy inputs with realistic error magnitudes, rather than validating an operational rolling forecasting system.

B.3. Error Distribution Fitting for Travel Time Prediction

Let $\mathcal{D}_{\text{tt}}$ denote the empirical dataset of link-level travel-time prediction errors extracted from App. B.2. Each observation corresponds to the difference between realized and predicted travel times over consecutive sensor pairs. To model the stochastic structure of these errors, we estimate their probability density using Kernel Density Estimation (KDE). Specifically, given $n = |\mathcal{D}_{\text{tt}}|$ samples, the KDE estimator is defined as

$$\hat{f}(\epsilon) = \frac{1}{nh} \sum_{a \in \mathcal{D}_{\text{tt}}} K\left(\frac{\epsilon - a}{h}\right), \tag{20}$$

where ϵ denotes the prediction-error random variable, a is an observed error sample, $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$ is the Gaussian kernel, and the bandwidth $h = 0.082$ is selected using Scott's rule. To assess the adequacy of KDE, we compare it with three alternative distributional models: Johnson SU, Gaussian Mixture Models (GMM), and the skew-normal family. Goodness-of-fit is evaluated using the Kolmogorov–Smirnov (KS) statistic and discrepancies in the first four moments (mean, standard deviation, skewness, and kurtosis) between empirical and synthetic samples. The results are summarized in Table 5. While GMM achieves the smallest KS statistic, KDE provides the best overall compromise between global fit and moment preservation. In particular, KDE exhibits negligible bias in the mean and standard deviation and substantially smaller discrepancies in skewness and kurtosis, indicating a more faithful representation of both the central mass and tail behavior of the empirical error distribution. zgThis moment-level fidelity is precisely

Method	KS Stat	Mean Diff	Std Diff	Skew Diff	Kurt Diff
KDE	0.042	**0.0001**	0.007	**0.254**	**1.793**
JOHNSON	0.022	0.005	0.051	1.620	208.981
GMM	**0.020**	0.002	**0.001**	2.440	51.339
SKEWNORM	0.211	0.018	0.019	5.452	87.860

Table 5
Distribution fitting comparison for travel-time prediction errors.

what matters for predict-then-optimize transport decisions: Peled et al. (2021) report that optimized performance is most sensitive to the skewness of the error distribution and to the presence of infrequent large errors (heavy tails), and that the squared error measure (RMSNE) is dominated by these rare large residuals. A parametric fit that matched the variance but flattened the skew or truncated the tail (as the skew-normal and, for kurtosis, the Johnson S_U fit do in Table 5) would therefore misrepresent exactly the features that drive the downstream optimization. The "Mean Diff" column also serves as our bias check (cf. App. B.2.1): the near-zero value indicates that the fitted residuals inherit the small, non-recentred bias of the trained DCRNN rather than an artificially imposed zero mean. Accordingly, we adopt the KDE-based estimator $\hat{f}$ in (20) to generate prediction-error samples ϵ_{ij}^{ω} used in the numerical experiments.

C. Algorithmic Framework and Evaluation

C.1. Parameter values for the proposed Algorithm 1

When computing adaptive network $\mathcal{G}_{\text{PT}}$ using the solution method described in § 4, we set the algorithmic parameters introduced therein as follows. The minimum path-overlap threshold is set to $J_{\min} = 0.5$ (§4.2.1), the crossover and mutation probabilities to $P_c = 0.85$ (§4.2.2) and $P_m = 0.35$ (§4.2.3), and the number of reference points to $N_{\text{ref}} = 5^3 = 125$ (§4.2.4), respectively.

C.2. Performance Metrics for Multi-Objective Optimization

We evaluate algorithm performance using three standard metrics for multi-objective optimization: Hypervolume (HV), Inverted Generational Distance (IGD), and Generational Distance (GD). The HV metric quantifies the volume of the objective space dominated by the Pareto solutions:

$$\text{HV}(X, P) = \Lambda\left(\bigcup_{x \in X} v(x, P)\right) \tag{21}$$

where X is the approximation set, P is the reference point, and $v(x, P)$ is the hypervolume dominated by solution x relative to P. Higher HV values indicate better coverage of the objective space. To ensure a fair comparison across algorithms, we report the HV metric calculated relative to a unified reference point. This reference point is constructed by collecting all solutions generated by all algorithms for each objective and selecting the worst (largest) value observed for each objective. The IGD metric measures the closeness of the approximation set to the true Pareto front:

$$\text{IGD}(X, \mathcal{P}^*) = \frac{\sum_{x^* \in \mathcal{P}^*} d(x^*, X)}{|\mathcal{P}^*|} \tag{22}$$

where $\mathcal{P}^*$ is a set of reference points on the true Pareto front, $d(x^*, X)$ is the minimum distance from x^* to any point in X, and $|\mathcal{P}^*|$ is the cardinality of $\mathcal{P}^*$. Lower IGD values indicate better convergence to the true Pareto front. The GD metric complements IGD by measuring how far the obtained solutions are from the true Pareto front:

$$\text{GD}(X, \mathcal{P}^*) = \frac{\sum_{x \in X} d(x, \mathcal{P}^*)}{|X|} \tag{23}$$

where $d(x, \mathcal{P}^*)$ is the minimum distance from a solution x to the true Pareto front $\mathcal{P}^*$, and $|X|$ is the cardinality of the approximation set. Lower GD values indicate better convergence of the approximation set to the true Pareto front. zgAs the true Pareto front $\mathcal{P}^*$ is unknown for the instances considered, we approximate it, following standard practice

in multi-objective optimization, by the non-dominated subset of the union of all solutions produced by all compared algorithms across all runs at a given variability level. The HV reference point is constructed analogously from the worst objective values observed across all algorithms (as described above), so that HV, IGD, and GD are all evaluated against a common, algorithm-agnostic reference.

C.3. Implementation Details of the NSGA-III Environmental Selection

The algorithm adopts the reference-point-based environmental selection mechanism of NSGA-III (Deb and Jain, 2014) to maintain diversity across the objectives (5). Since all objectives are formulated as minimization problems, normalization is performed on the combined population $R_t = P_t \cup Q_t$. For each objective $j \in \{1, 2, 3\}$, objective-wise lower and upper bounds are computed as $z_j^{\min} = \min_{x \in R_t} Z_j(x)$ and $z_j^{\max} = \max_{x \in R_t} Z_j(x)$, and each objective is linearly scaled to $[0, 1]$ accordingly.

A set of uniformly distributed reference points on the unit simplex is generated following the simplex-lattice design in Deb and Jain (2014). As in standard multiobjective optimization, we say that solution x dominates y if x is no worse than y in all objectives and strictly better in at least one. We employ non-dominated sorting, which partitions the **combined population** $R_t = P_t \cup Q_t$ (comprising both parent and offspring individuals) into a sequence of Pareto fronts $F = \{F_1, F_2, \ldots, F_l\}$. Here, F_1 contains the globally non-dominated solutions F_2 contains solutions dominated exclusively by members of F_1, and F_l represents the *critical front* where the cumulative count of solutions first meets or exceeds the population capacity *PopSize*. Each solution is associated with its nearest reference point by computing perpendicular distances from the solution to the reference lines in the normalized space. The environmental selection process (line 10 of Alg. 1) adds complete Pareto fronts F_1 through F_{l-1} sequentially; for the last front F_l, slots are filled by prioritizing solutions associated with reference points that have fewer assigned solutions to ensure diversity.

Algorithm 2 Feasibility-Oriented Path Completion for Line Repair

Input: Substrate graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, endpoints (u, v), visited node set $\mathcal{V}^{\ell}_{\text{used}}$
Output: Connecting path $\mathcal{P}_{u \to v}$ or $\emptyset$

$visited \leftarrow \{u\} \cup \mathcal{V}^{\ell}_{\text{used}}$, $queue \leftarrow \langle u \rangle$, $parent(\cdot) \leftarrow \emptyset$
while *queue* is not empty **do**
  $i \leftarrow queue$.pop_front()
  if $i = v$ **then**
    break
  end if
  for each neighbor j of i in $\mathcal{G}$ **do**
    if $j \notin visited$ **then**
      $visited \leftarrow visited \cup \{j\}$
      $parent(j) \leftarrow i$
      $queue$.push_back(j)
    end if
  end for
end while
if $v \notin visited$ **then**
  return $\emptyset$
end if
$\mathcal{P}_{u \to v} \leftarrow$ Backtrack from v to u using $parent(\cdot)$
return $\mathcal{P}_{u \to v}$

C.4. Performance of NSGA-III for Static Optimal Network Generation

For generating our baseline static optimal network $\mathcal{G}^{\text{static}}_{\text{PT}}$, we employ NSGA-III for multi-objective optimization . Fig. 6 displays the resulting Pareto frontier with non-dominated solutions balancing vehicle operation time against passenger travel time. We identify $\mathcal{G}^{\text{static}}_{\text{PT}}$ as the solution that is closest to the ideal point, defined as the point where all objectives attain their minimum value. Specifically, we select the Pareto-optimal solution that minimizes the normalized Euclidean distance to this ideal point. This choice corresponds to the "knee" region of the Pareto frontier, where any further improvement in one objective would cause a disproportionately large deterioration in the other.

Performance Metric	NSGA-II	NSGA-III	NSVND-II	NSVND-III
Inverted Generational Distance (IGD, lower is better)	347.64	**22.63**	38.90	34.80
IGD Change[a]	–	**-325.01**	-308.74	-312.84
Generational Distance (GD, lower is better)	75.36	**17.17**	24.40	22.80
GD Change[a]	–	**-58.19**	-50.96	-52.56
Hypervolume (HV, higher is better) [b]	30.52×10^9	31.49×10^9	31.23×10^9	31.31×10^9
HV Change[a]	–	**0.97**$\times 10^9$	0.71$\times 10^9$	0.79$\times 10^9$

[a] Improvement/Change are computed relative to NSGA-II (baseline).
[b] HV uses a single fixed reference point shared by all algorithms.

Table 6
Multi-objective algorithm comparison under noise-free travel times ($cv = 0$).

C.5. Algorithm Performance Evaluation

Algorithmic performance is evaluated via the Hypervolume (HV), the Inverted Generational Distance (IGD), and the Generational Distance (GD), which are standard performance metrics in multi-objective evolutionary optimization as surveyed in (Riquelme, Von Lücken and Baran, 2015).[6] The Hypervolume (HV) evaluates both convergence and diversity by measuring the dominated volume in the objective space, where higher values indicate better performance (↑). In contrast, GD and IGD are distance-based indicators that assess convergence to a reference Pareto front, with lower values indicating better convergence (↓).

To evaluate the effectiveness of NSGA-III for zg~~real-time~~forecast-triggered PT redesign, we compare it with NSGA-II (Deb, Pratap, Agarwal and Meyarivan, 2002) and hybrid multi-objective algorithms that combine non-dominated sorting with variable neighborhood descent, based on Mandl's Swiss network. Specifically, we consider two Non-dominated Sorting Variable Neighborhood Descent (NSVND) variants inspired by the integration of NSGA-II with variable neighborhood search strategies (Hansen and Mladenović, 2001). The first variant (NSVND-II) follows the NSGA-II framework while replacing crossover and mutation with variable neighborhood descent, whereas the second variant (NSVND-III) further incorporates an adaptive neighborhood selection mechanism to enhance exploration of the Pareto frontier.

Table 6 presents an algorithm performance comparison for adaptive network optimization assuming no travel time deviation from the nominal ones (cv=0). Although no adaptive redesign is required when $cv = 0$, this setting provides a noise–free benchmark for comparing optimization algorithms. NSGA-III maintains an optimal balance between solution diversity (high hypervolume) and convergence precision.

D. Network Performance Metrics

This appendix details the performance indicators used in § 6.1.3. For each realization $\omega \in \Omega$ of stochastic travel times, we define the following performance metrics: **20% Improvement Share (IS_{20})**, **Operational Cost Reduction (OCR)**, and **Route Overlap (RO)**. Since these metrics vary with the realized scenario ω, they are treated as random variables. In our numerical results, we report their empirical averages (and zg~~confidence intervals) over five~~standard deviations) over ten independent realizations.

- **20% Improvement Share (%):** The proportion of OD pairs experiencing a travel time reduction of at least 20% under realization ω.

$$\mathrm{IS}_{20}(\omega) = \frac{\sum_{k \in \mathcal{K}} \mathbb{I}\left(\frac{t_k^0 - t_k^*(\omega)}{t_k^0} \geq 20\%\right)}{|\mathcal{K}|} \times 100\%, \text{ where } \mathbb{I}(\cdot) \text{ is the indicator function.}$$

  Here t_k^0 and $t_k^*(\omega)$ denote the original and adaptive travel times for OD pair k, respectively. Since both depend on random travel-time realizations, IS_{20} is a random variable.

[6] Formal definitions and computational details are provided in App. C.2.

- **Operational Cost Reduction (OCR)**

$$\mathrm{OCR}(\omega) = \frac{\sum_{\ell \in \mathcal{L}} n^{\ell} \cdot t^{\ell} - \sum_{\ell \in \mathcal{L}} n^{\ell,*}(\omega) \cdot t^{\ell,*}(\omega)}{\sum_{\ell \in \mathcal{L}} n^{\ell} \cdot t^{\ell}} \times 100\%.$$

  Here t^{ℓ} and $t^{\ell,*}(\omega)$ are the realized travel times of route ℓ in the original and adaptive networks, respectively, and n^{ℓ} and $n^{\ell,*}$ are the corresponding vehicle allocations. Because t^{ℓ} and $t^{\ell,*}(\omega)$ vary with the realization, OCR is also a random variable.

- **Route Overlap (RO)** To quantify structural similarity between the original and adaptive networks, we use the Jaccard index:

$$\mathrm{RO}(\omega) = \frac{|\mathcal{E}_{\mathrm{PT}} \cap \mathcal{E}^{*}_{\mathrm{PT}}(\omega)|}{|\mathcal{E}_{\mathrm{PT}} \cup \mathcal{E}^{*}_{\mathrm{PT}}(\omega)|} \times 100\%.$$

  Here $\mathcal{E}_{\mathrm{PT}}$ and $\mathcal{E}^{*}_{\mathrm{PT}}(\omega)$ denote the realized edge sets of the original and adaptive networks, respectively. Since these sets depend on the travel-time realization, RO is likewise a random variable.

zg~~The numerical results therefore report the empirical means (and 95% confidence intervals) of PTI, OCR, and RO over the five realizations $\omega \in \Omega$, providing a stochastic assessment of adaptive-network performance.~~The numerical results therefore report the empirical means $\pm$ standard deviations of IS_{20}, OCR, and RO over the ten realizations $\omega \in \Omega$, providing a stochastic assessment of adaptive-network performance.

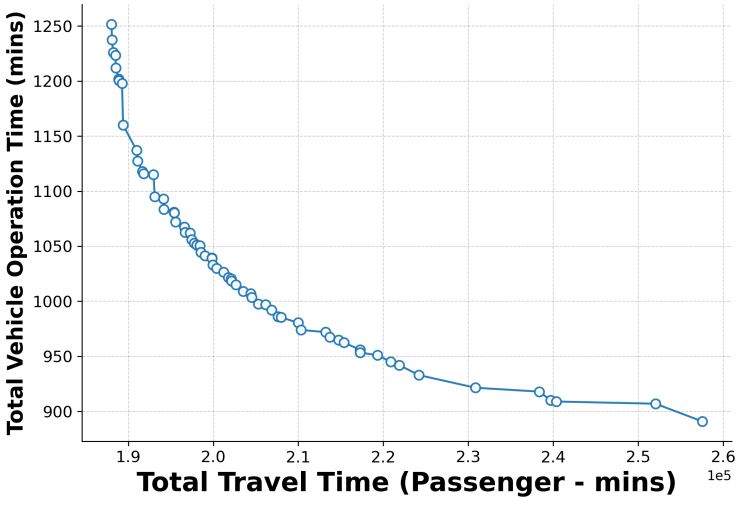


Figure 6: Pareto Frontier Generated by NSGA-III for Static Optimal Network